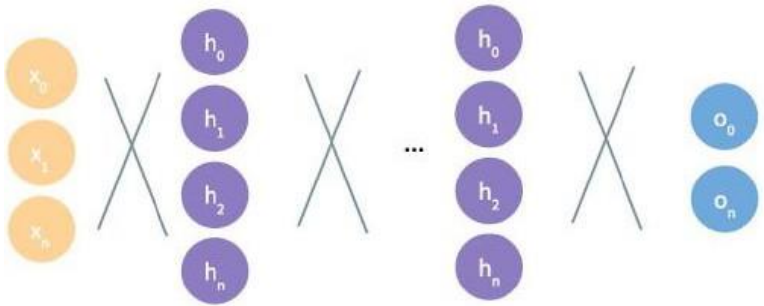
Intro to Deep Learning

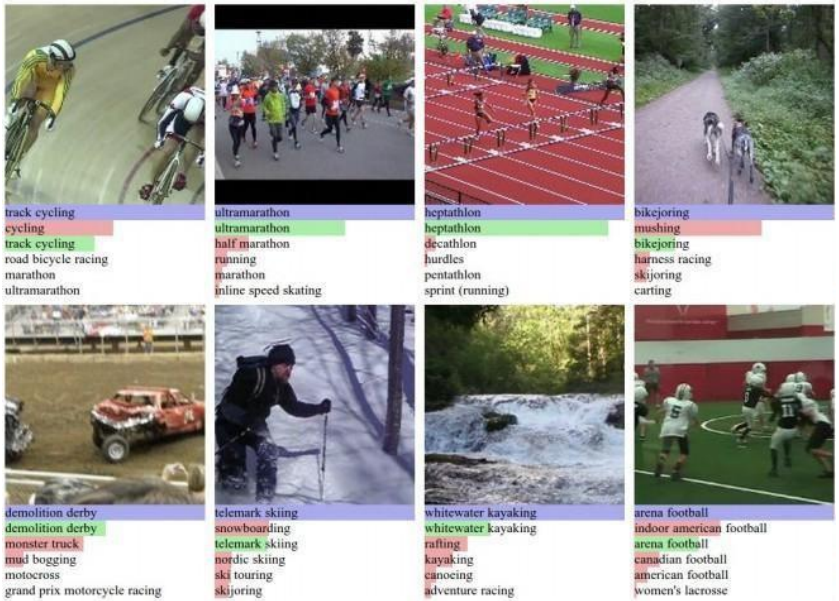
Kheireddin Kadri



Deep Learning Success

- Image Classification
Machine Translation
Speech Recognition
Speech Synthesis
Game Playing

… and many, many more



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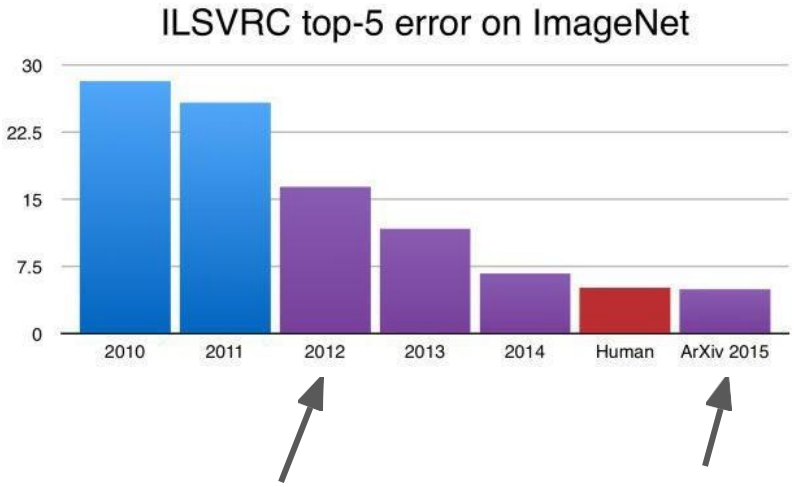


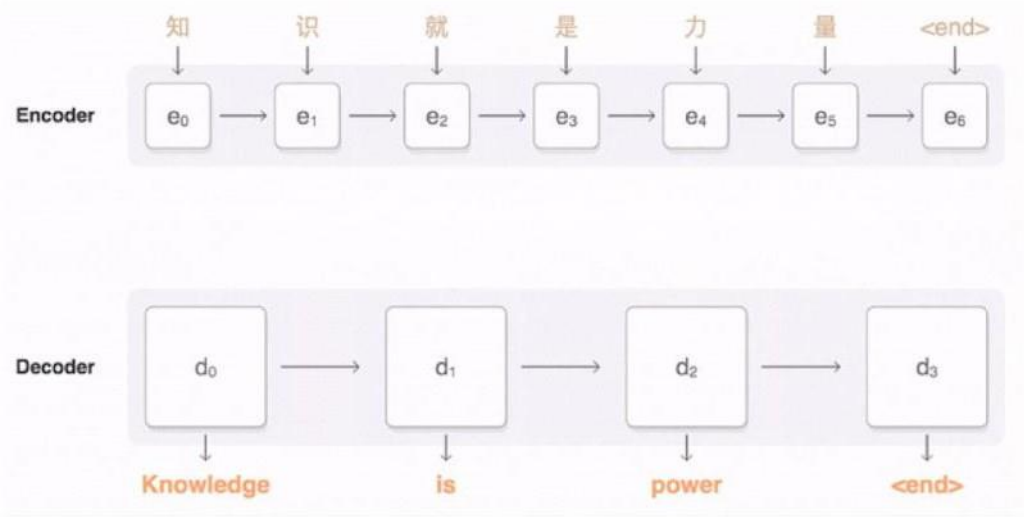
Krizhevsky, Sutskever, Hinton 2012

Deep Learning Success

Google Translate

Image Classification
- Machine Translation
Speech Recognition
Speech Synthesis
Game Playing

.
… and many, many more



Deep Learning Success

Image Classification
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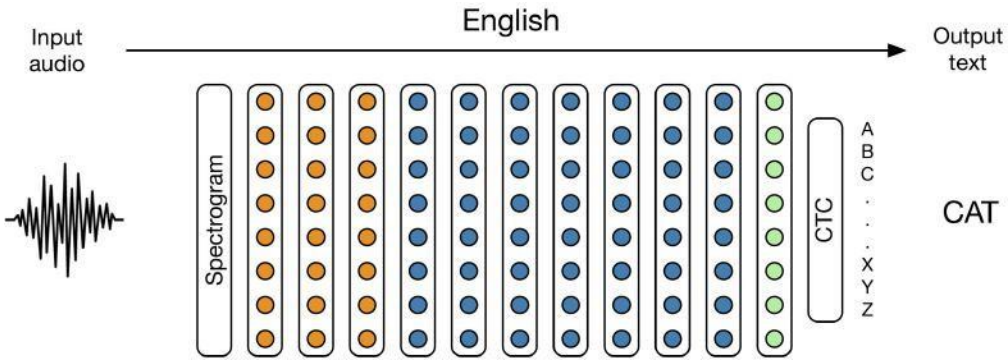
… and many, many more



Deep Learning Success

Image Classification
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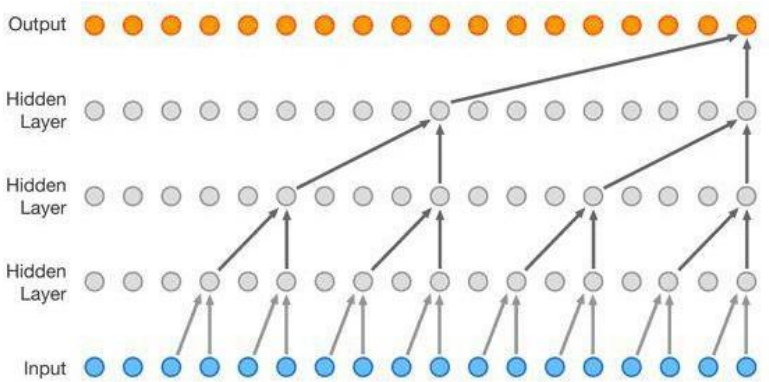
… and many, many more



Deep Learning Success

Image Classification
Machine Translation
Speech Recognition
Speech Synthesis
- Game Playing

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… and many, many more

Deep Learning Success

Image Classification
Machine Translation
Speech Recognition
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.
… and many, many more

Goals

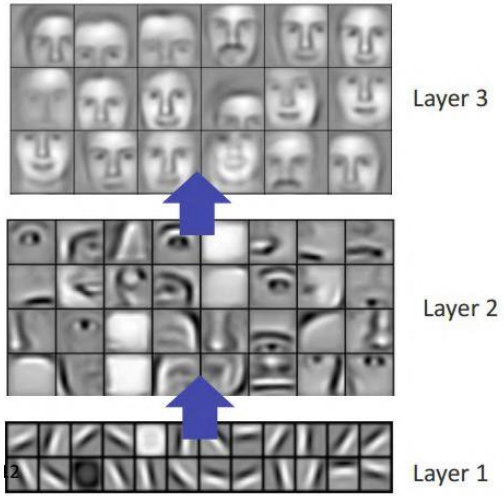
1. Fundamentals
2. Practical skills
3. Up to speed on current state of the field

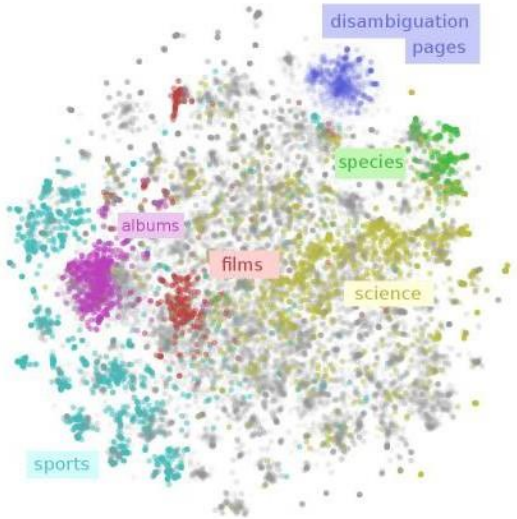
Knowledge, intuition, know-how, and community to do deep learning research and development.

Why Deep Learning and why now?

Why Deep Learning?

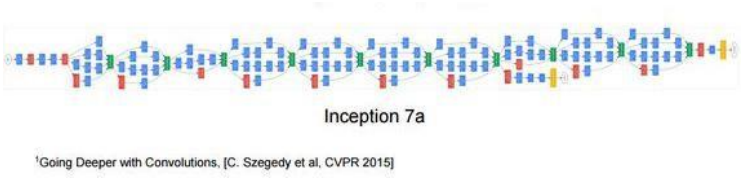
- Hand-Engineered Features vs. Learned features



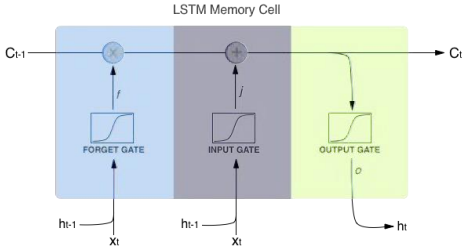


Why Now?

1. Large Datasets
2. GPU Hardware Advances + Price Decreases
3. Improved Techniques



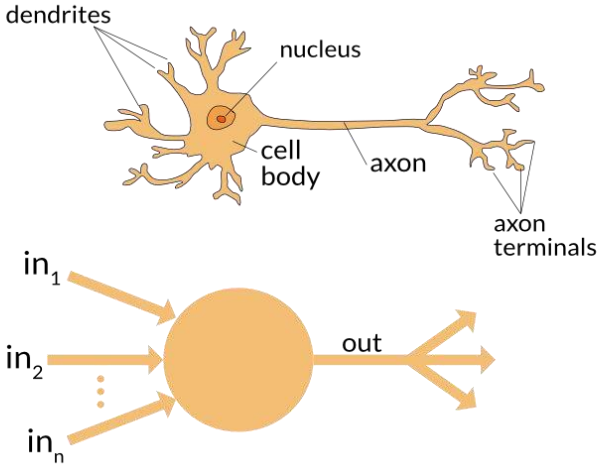
[1]Going Deeper with Convolutions, [C. Szegedy et al, CVPR 2015]



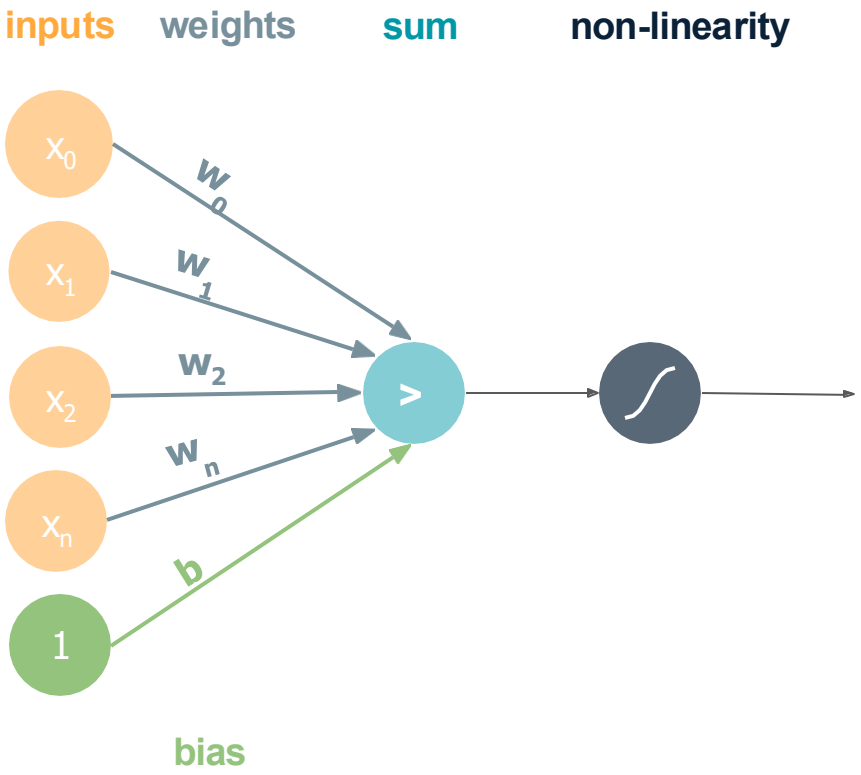
Fundamentals of Deep Learning

The Perceptron

1. Invented in 1954 by Frank Rosenblatt
2. Inspired by neurobiology

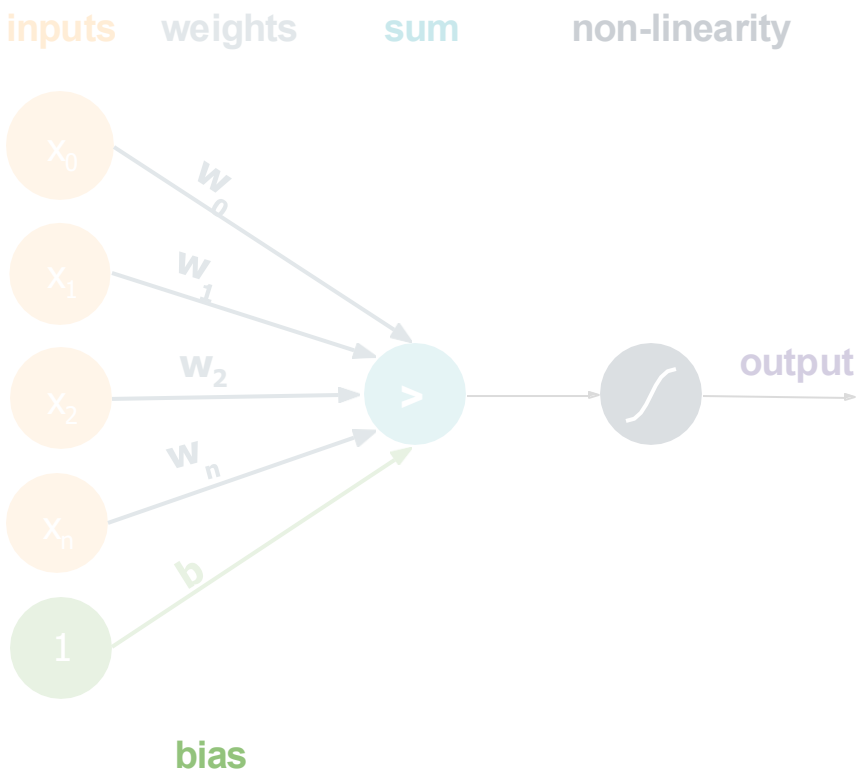


The Perceptron



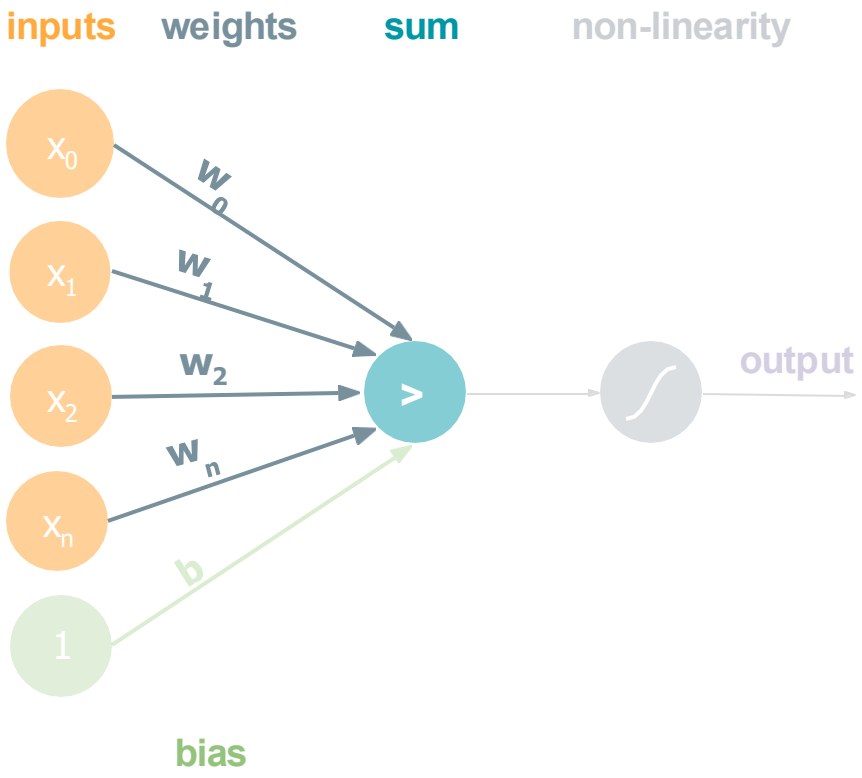
Perceptron Forward Pass

$$output =$$



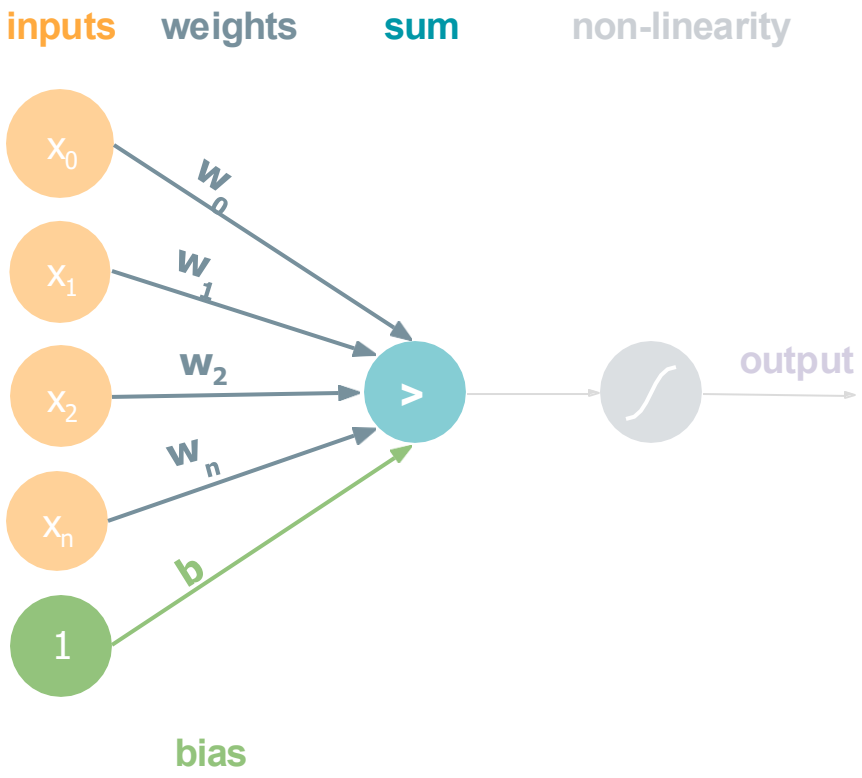
Perceptron Forward Pass

$$output = \sum_{i=0}^{N} x_i * w_i$$



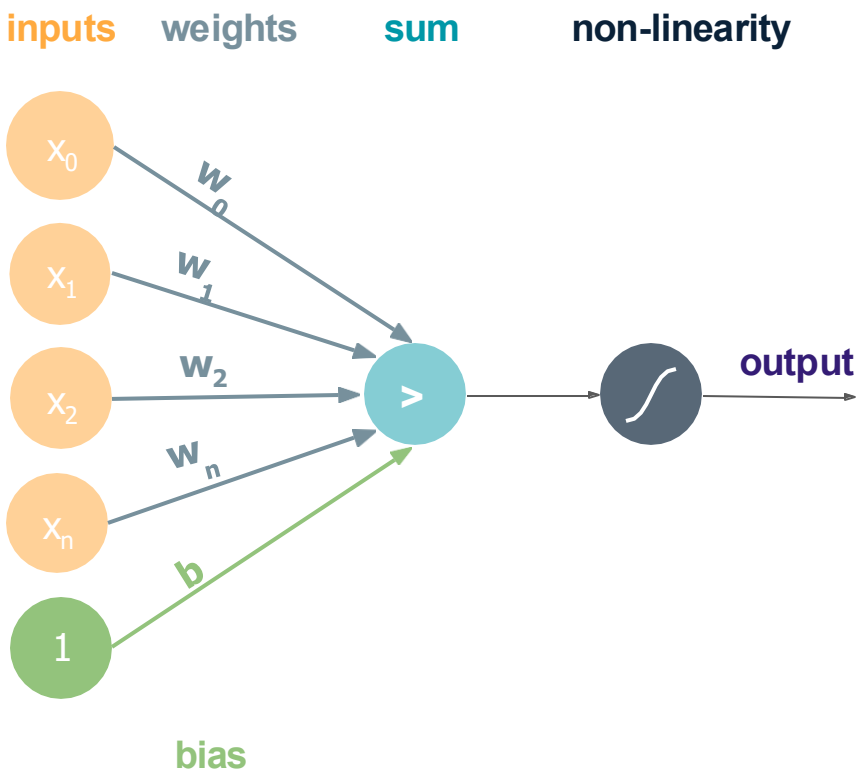
Perceptron Forward Pass

$$output = \sum_{i=0}^{N} (x_i * w_i) + b$$



Perceptron Forward Pass

$$output = g((\sum_{i=0}^{N} x_i * w_i) + b)$$

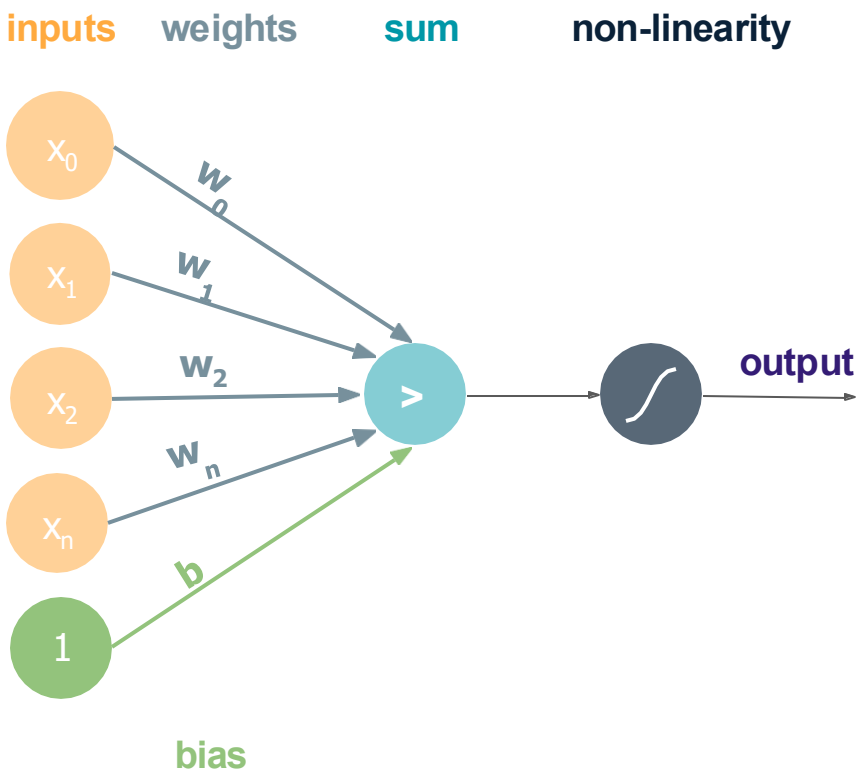


Perceptron Forward Pass

$$output = g(XW + b)$$

$$X = x_0, x_1, ...x_n$$

$$W = w_0, w_1, ...w_n$$



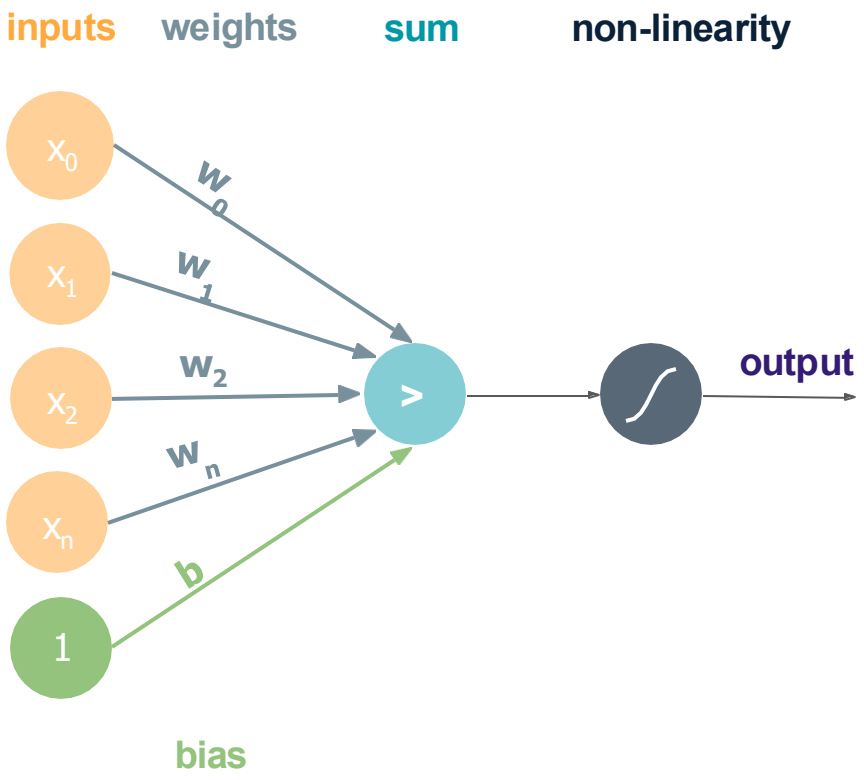
Perceptron Forward Pass

Activation Function

$$output = g(XW + b)$$

$$X = x_0, x_1, ...x_n$$

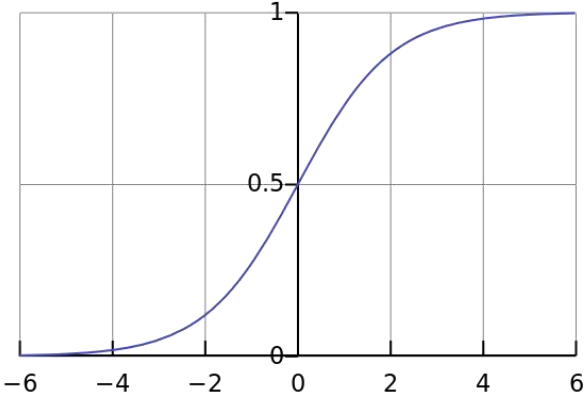
$$W = w_0, w_1, ...w_n$$

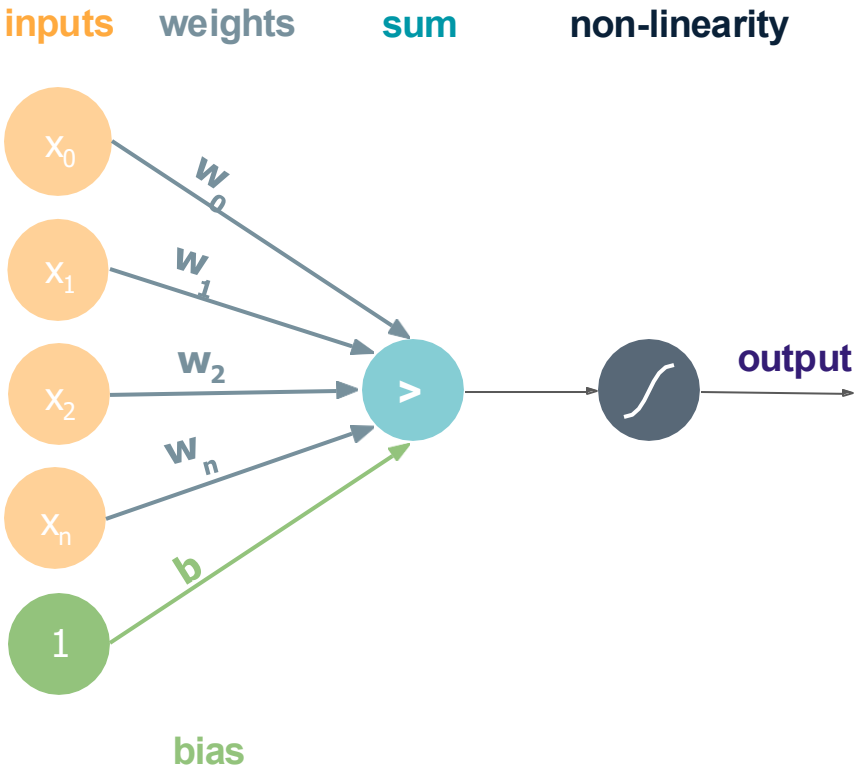


Sigmoid Activation

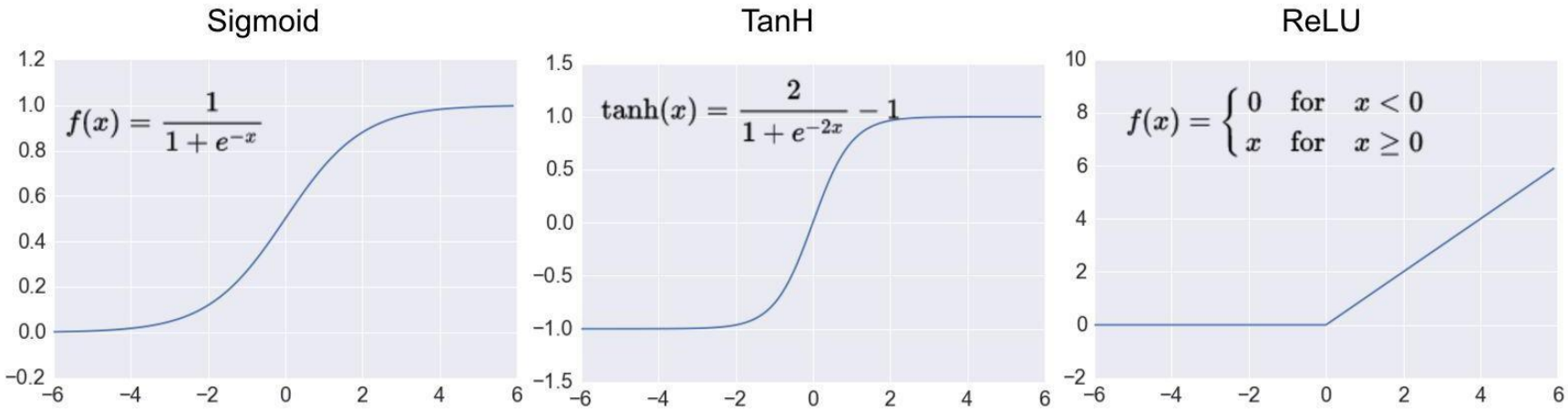
$$output = g(XW + b)$$

$$g(a) = \sigma(a) = \frac{1}{1 + e^{-a}}$$



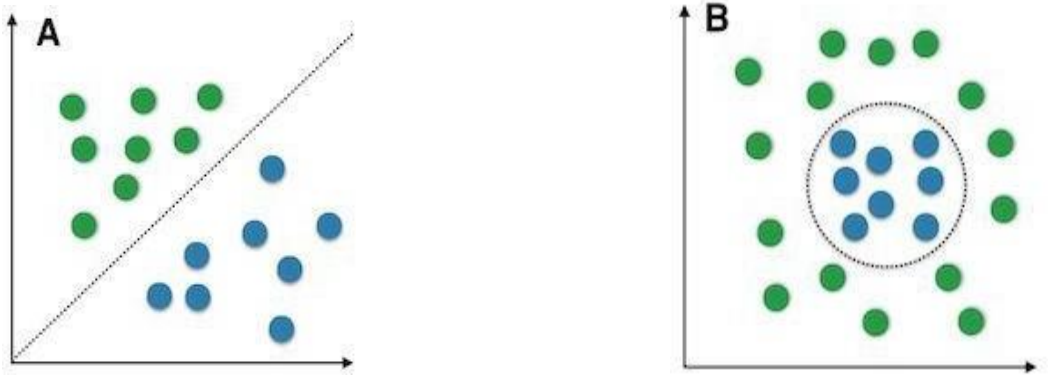


Common Activation Functions



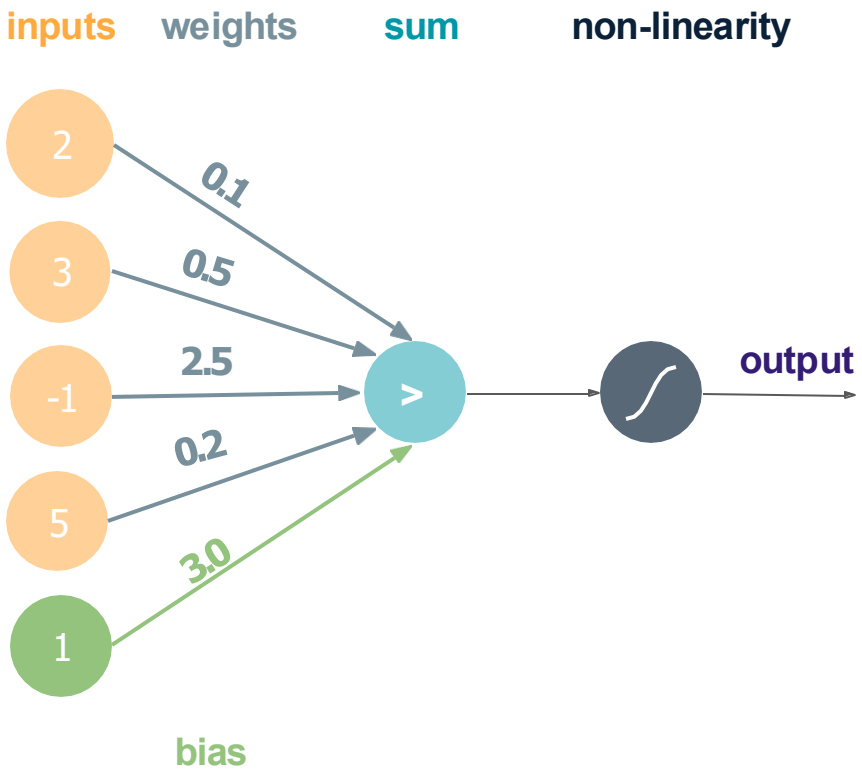
Importance of Activation Functions

- Activation functions add non-linearity to our network's function
- Most real-world problems + data are **non-linear**



Perceptron Forward Pass

$$output = g(XW + b)$$



Perceptron Forward Pass

$$output = g($$

(2*0.1) +

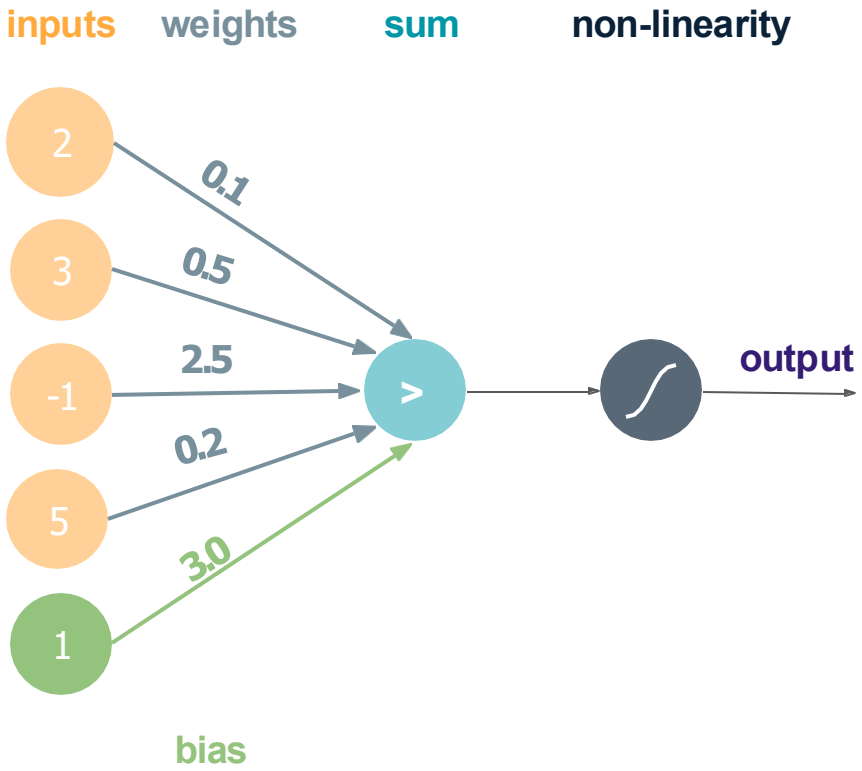
(3*0.5) +

(-1*2.5) +

(5*0.2) +

(1*3.0)

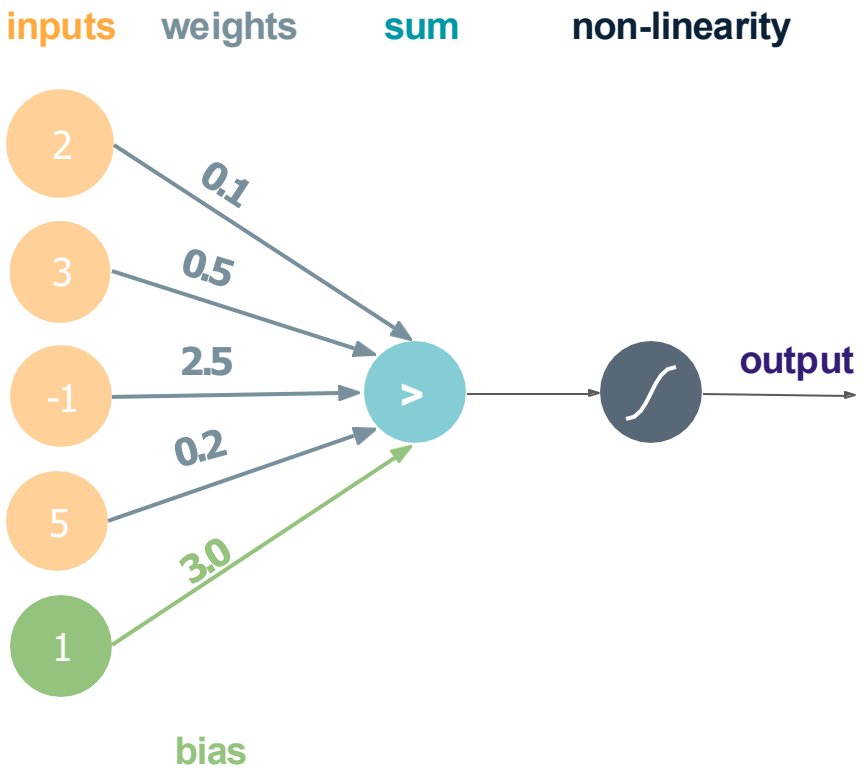
)



Perceptron Forward Pass

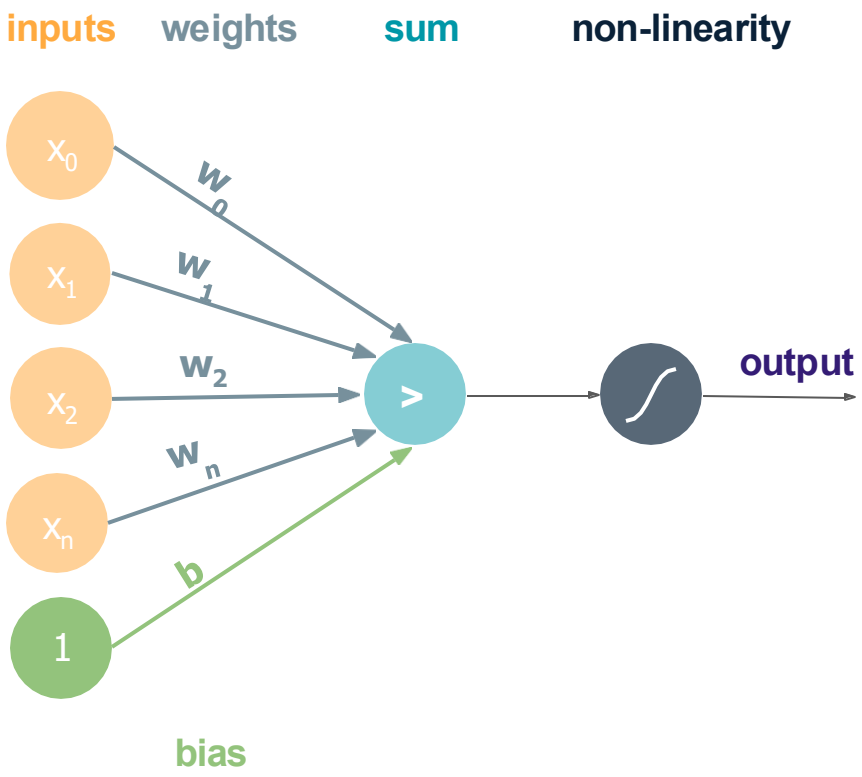
$$output = g(3.2) = \sigma(3.2)$$

$$= \frac{1}{(1 + e^{-3.2})} = 0.96$$

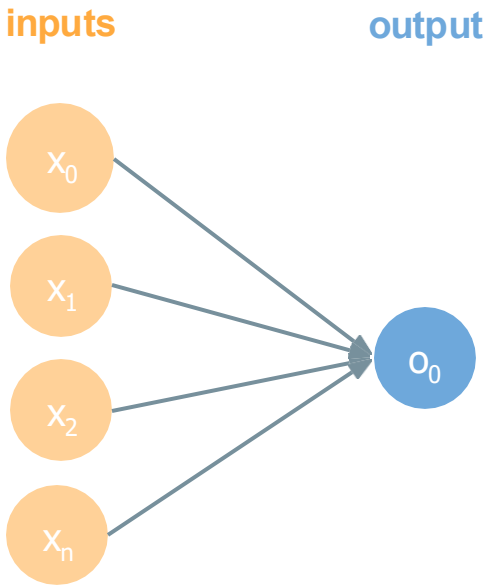


How do we build neural networks with perceptrons?

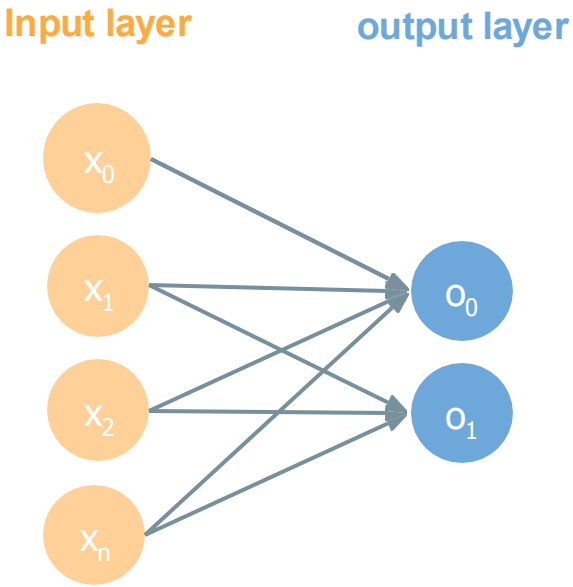
Perceptron Diagram Simplified



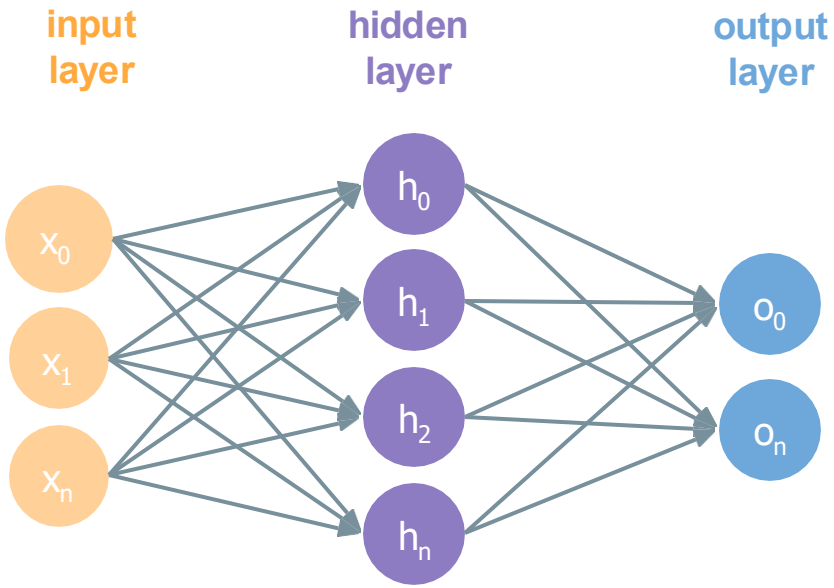
Perceptron Diagram Simplified



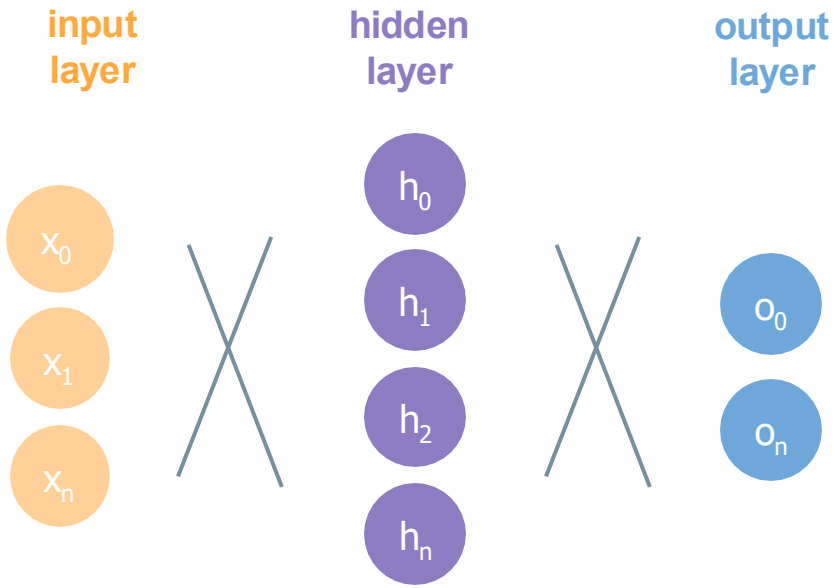
Multi-Output Perceptron



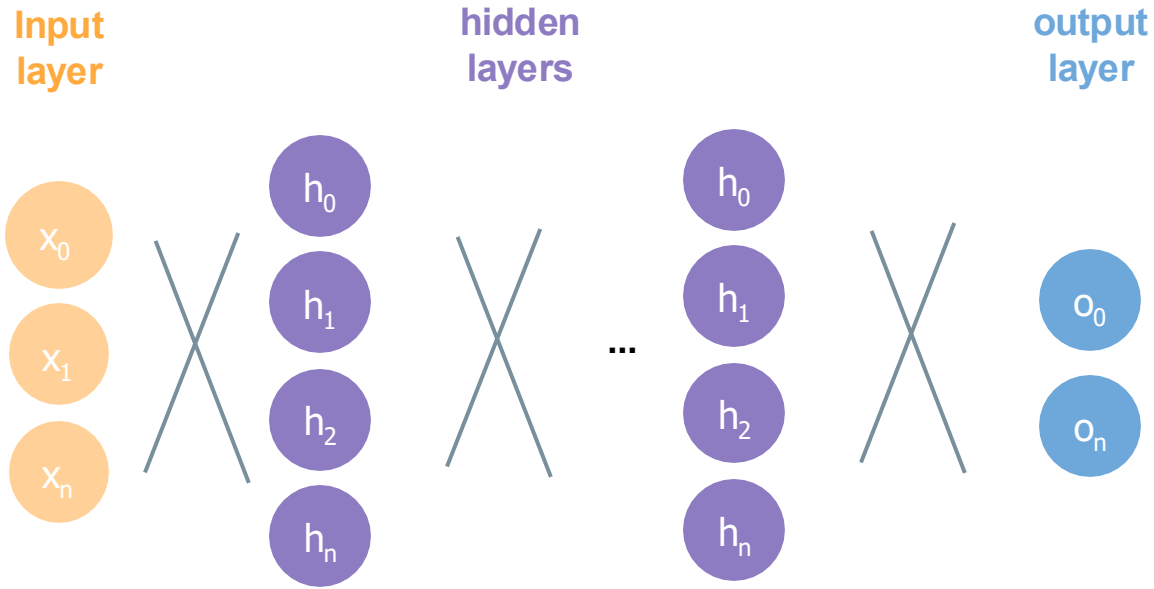
Multi-Layer Perceptron (MLP)



Multi-Layer Perceptron (MLP)



Deep Neural Network



Applying Neural Networks

Example Problem: Will my Flight be Delayed?

Example Problem: Will my Flight be Delayed?

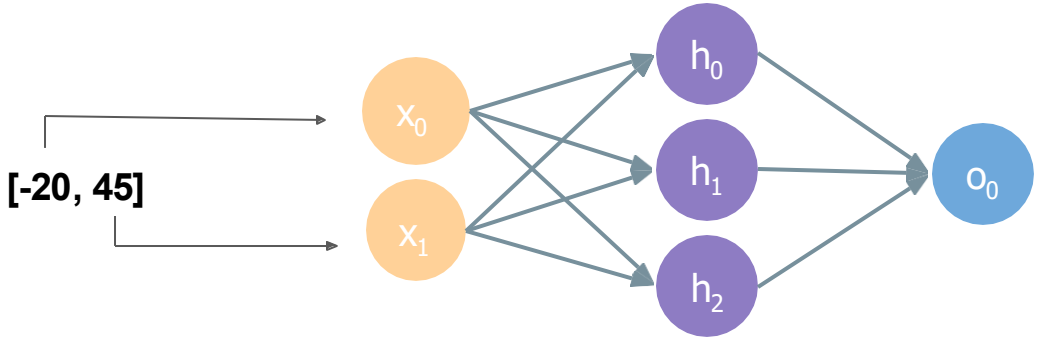
Temperature: -20 F

Wind Speed: 45 mph

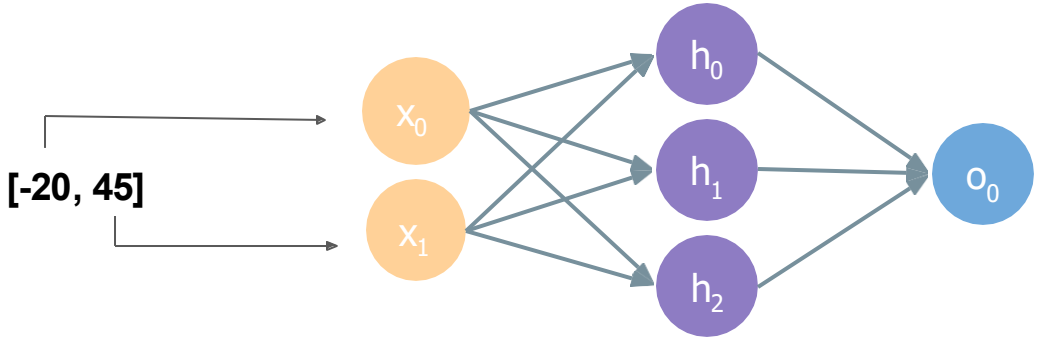
Example Problem: Will my Flight be Delayed?

[-20, 45]

Example Problem: Will my Flight be Delayed?

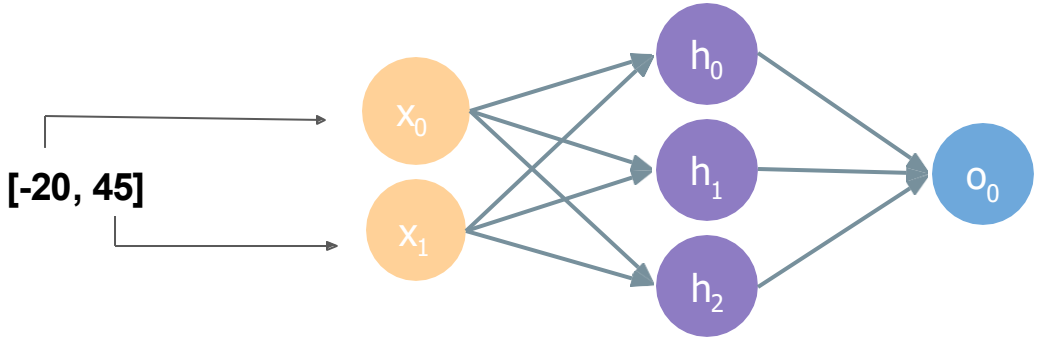


Example Problem: Will my Flight be Delayed?



Predicted: 0.05

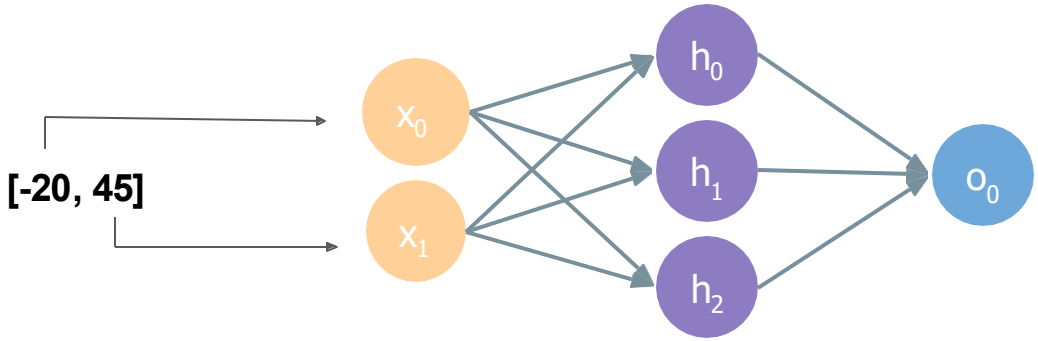
Example Problem: Will my Flight be Delayed?



Predicted: 0.05

Actual: 1

Quantifying Loss



Predicted: 0.05

Actual: 1

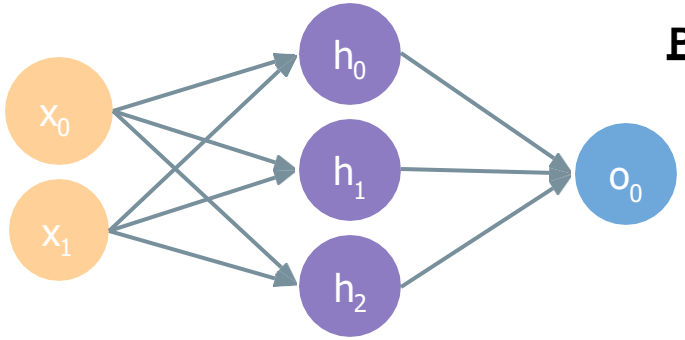
$$loss(f(x^{(i)};\theta), y^{(i)})$$

Predicted Actual

Total Loss

Input

[
[-20, 45],
[80, 0],
[4, 15],
[45, 60],
]



Predicted	**Actual**
[	[
0.05	1
0.02	0
0.96	1
0.35	1
]	]

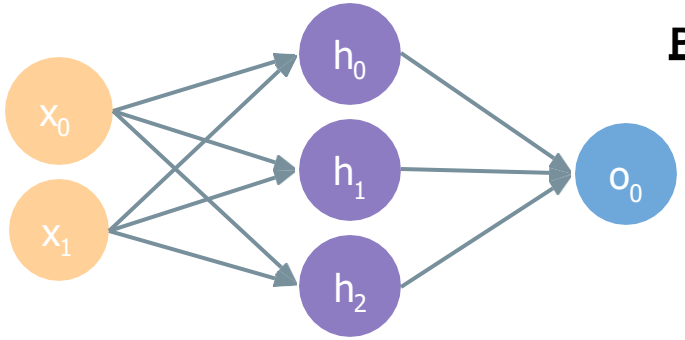
$$\text{total loss} := J(\theta) = \frac{1}{N}\sum_{i}^{N} loss(f(x^{(i)};\theta), y^{(i)}))$$

Predicted Actual

Total Loss

Input

[
[-20, 45],
[80, 0],
[4, 15],
[45, 60],
]



Predicted	**Actual**
[	[
0.05	1
0.02	0
0.96	1
0.35	1
]	]

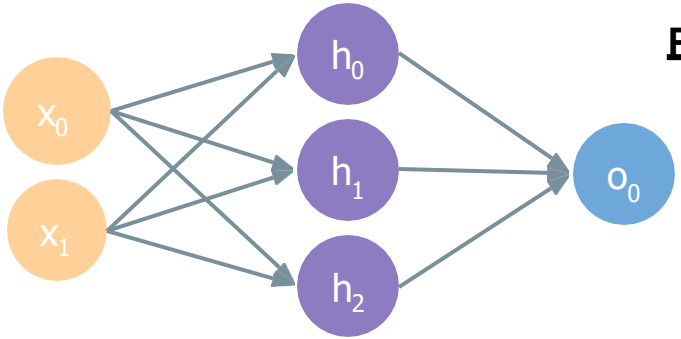
$$\text{total loss} := J(\theta) = \frac{1}{N}\sum_{i}^{N} loss(f(x^{(i)};\theta), y^{(i)}))$$

Predicted Actual

Binary Cross Entropy Loss

Input

[
[-20, 45],
[80, 0],
[4, 15],
[45, 60],
]



Predicted	**Actual**
[	[
0.05	1
0.02	0
0.96	1
0.35	1
]	]

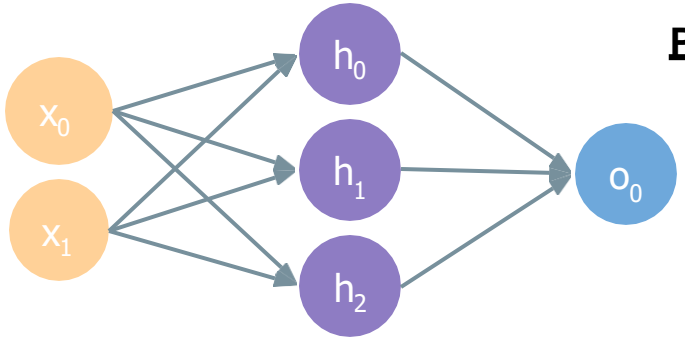
$$\text{cross\_entropy}(\theta) = \frac{1}{N}\sum_{i}^{N} y^{(i)} log(f(x^{(i)};\theta)) + (1 - y^{(i)}) log(1 - f(x^{(i)};\theta)))$$

Actual Predicted Actual Predicted

Mean Squared Error (MSE) Loss

Input

[
[-20, 45],
[80, 0],
[4, 15],
[45, 60],
]



Predicted	**Actual**
[	[
10	40
45	42
100	110
15	55
]	]

$$\mathrm{MSE}(\theta) = \frac{1}{N}\sum_{i}^{N}(f(x^{(i)};\theta) - y^{(i)})^2$$

Predicted Actual

Training Neural Networks

Training Neural Networks: Objective

$$arg_\theta \min \frac{1}{N} \sum_{i}^{N} \text{loss}(f(x^{(i)};\theta), y^{(i)})$$

Training Neural Networks: Objective

$$arg_\theta \min \underbrace{\frac{1}{N}\sum_{i}^{N} \text{loss}(f(x^{(i)};\theta), y^{(i)})}_{J(\theta)}$$

loss function

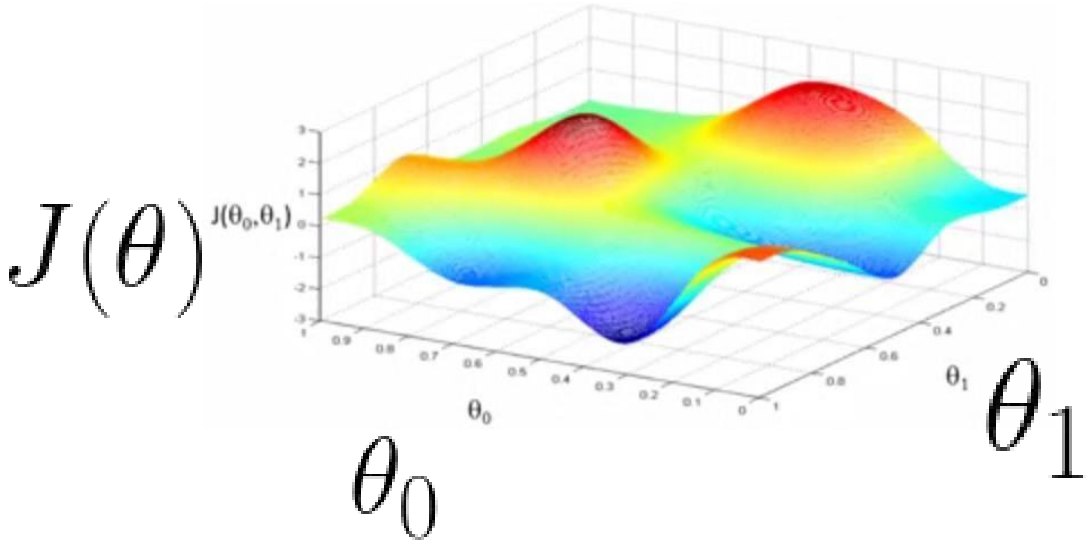
Training Neural Networks: Objective

$$arg_\theta \min \frac{1}{N} \sum_i^N \text{loss}(f(x^{(i)};\theta), y^{(i)})$$

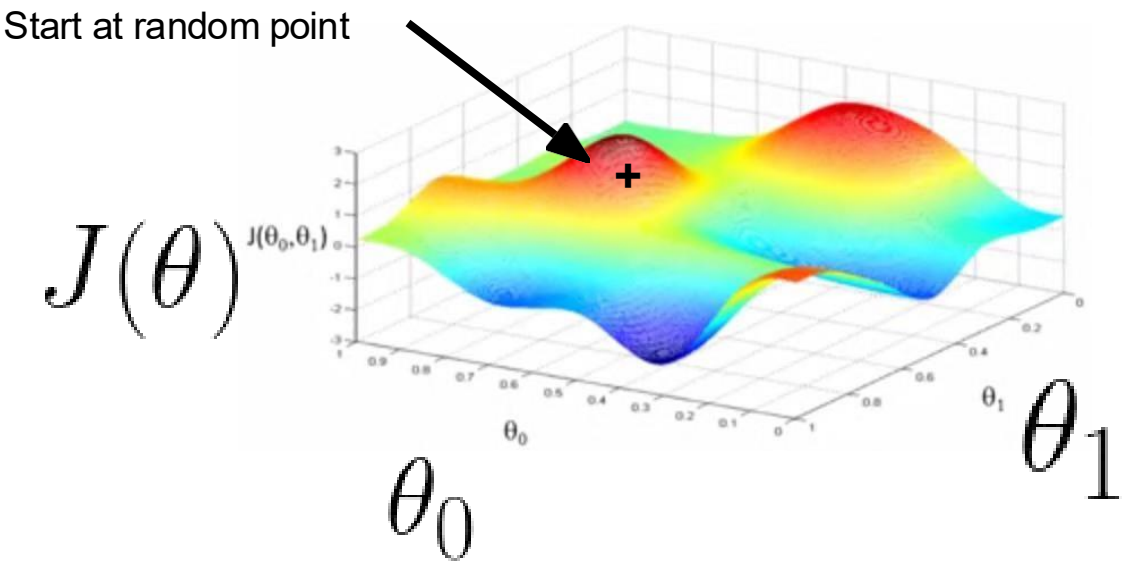
$$J(\theta)$$

$$\theta = W_1, W_2 ... W_n$$

Loss is a **function** of the model's parameters

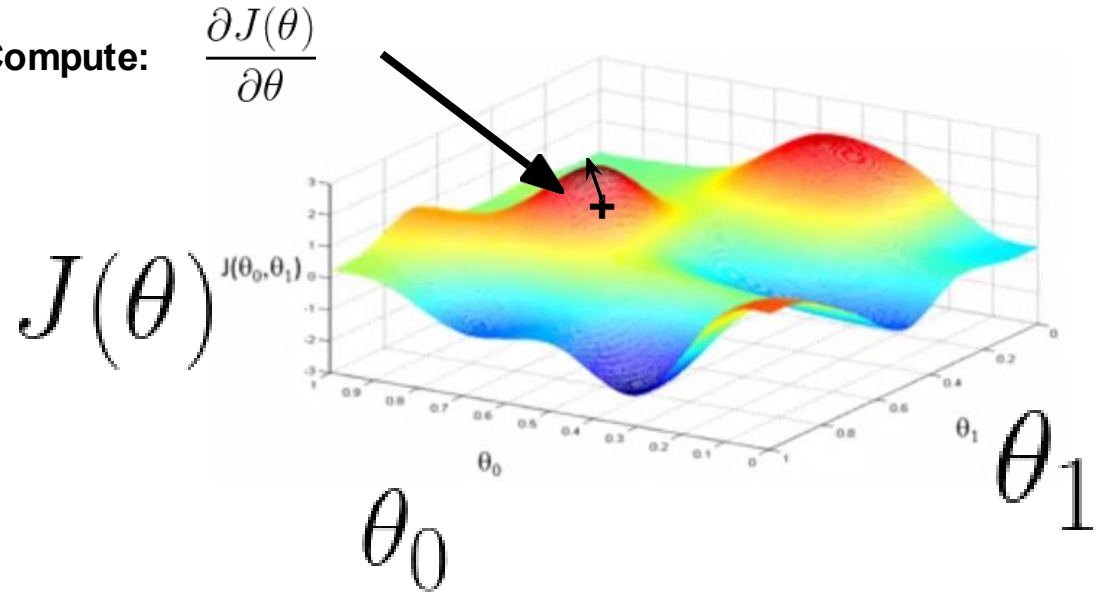


How to minimize loss?



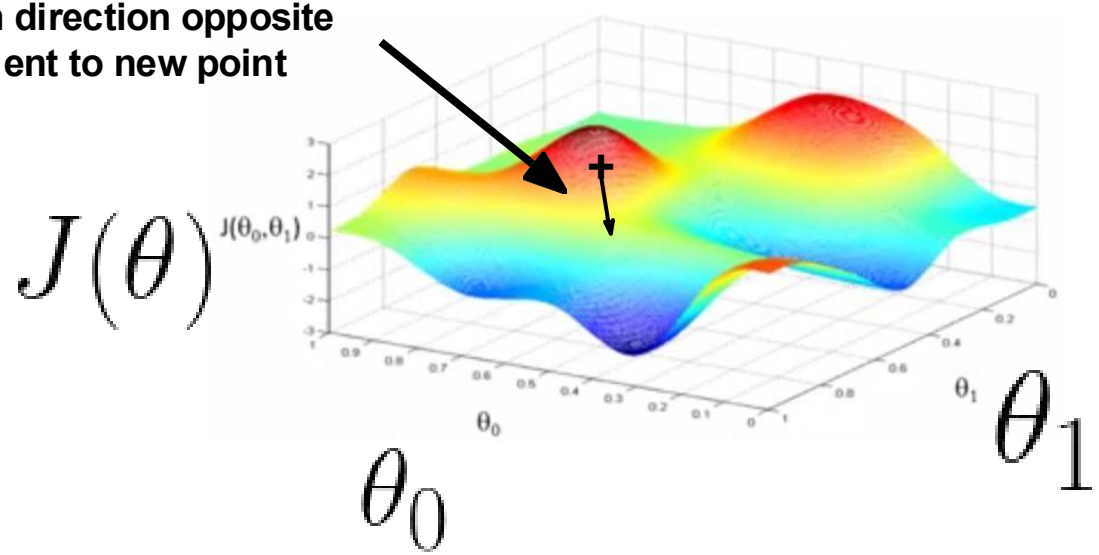
How to minimize loss?

Compute: $\frac{\partial J(\theta)}{\partial \theta}$



How to minimize loss?

Move in direction opposite of gradient to new point



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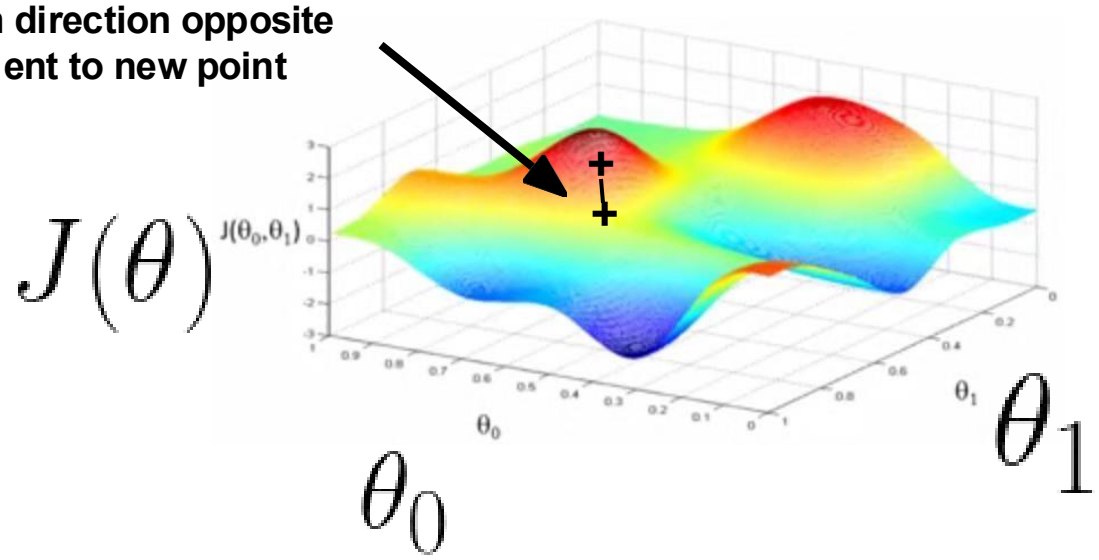
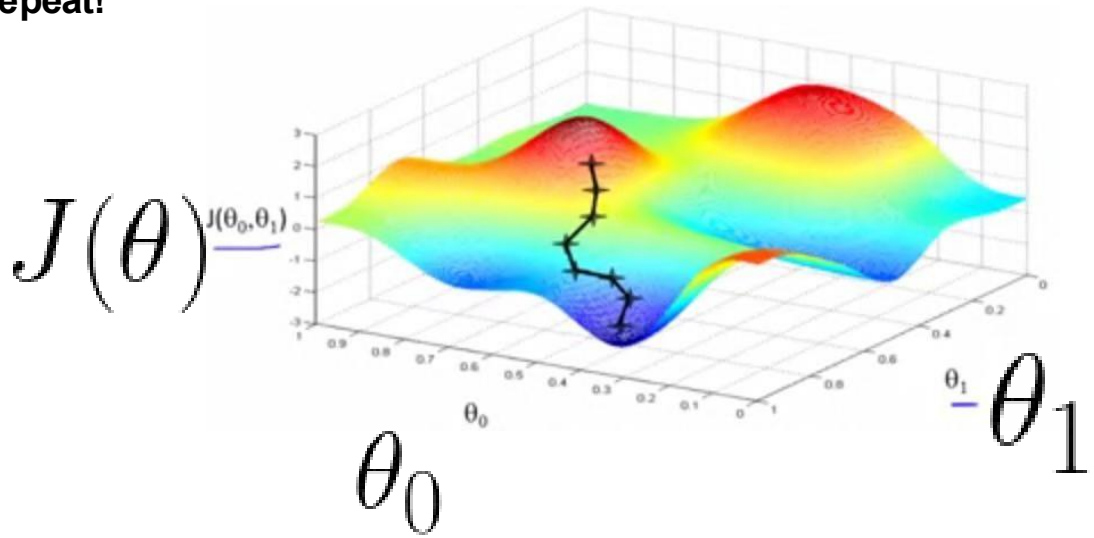


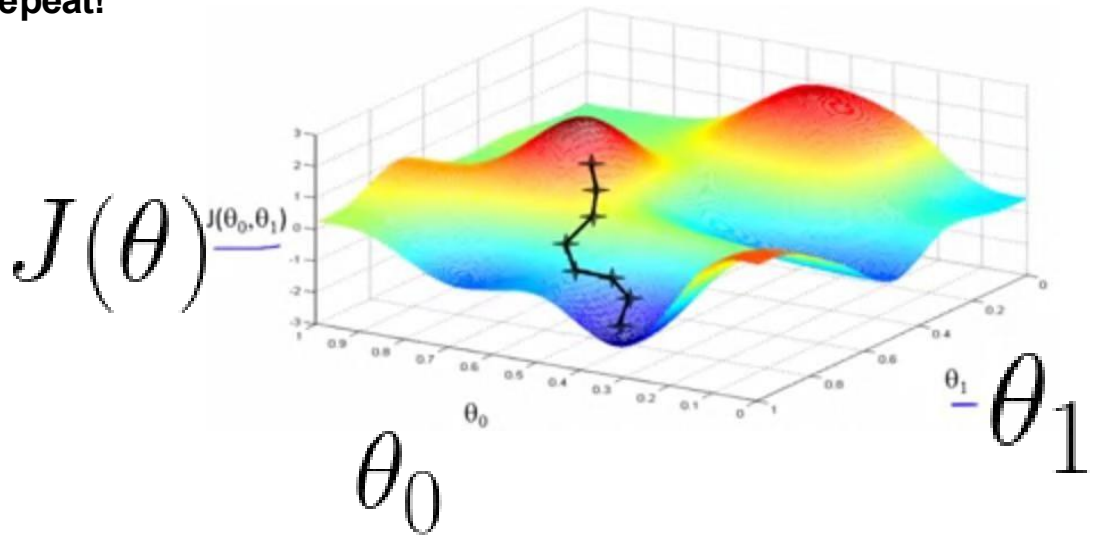
How to minimize loss?

Repeat!



This is called Stochastic Gradient Descent (SGD)

Repeat!



Stochastic Gradient Descent (SGD)

- Initialize θ randomly
- For N Epochs
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    - Compute Loss Gradient: $\frac{\partial J(\theta)}{\partial \theta}$
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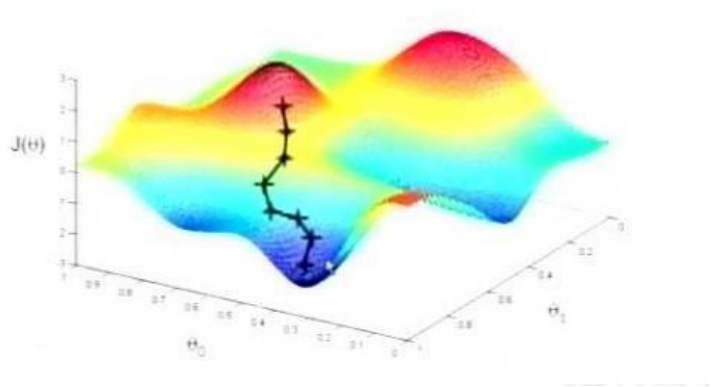
$$\theta := \theta - \eta \frac{\partial J(\theta)}{\partial \theta}$$



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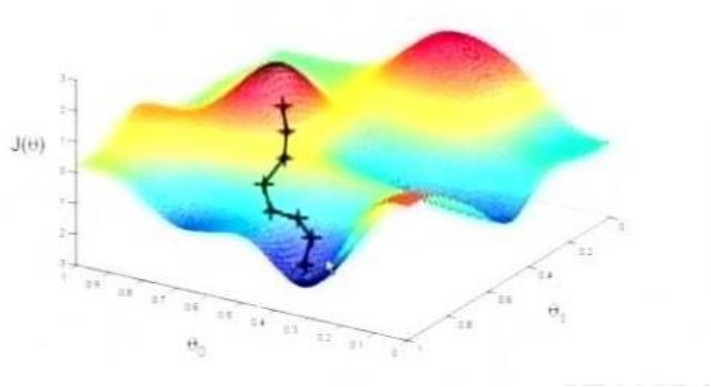

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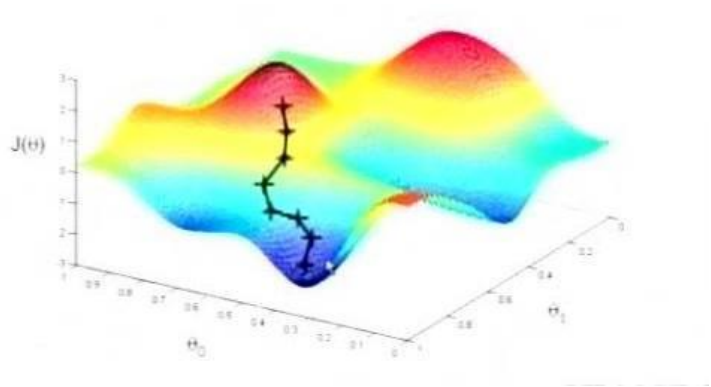
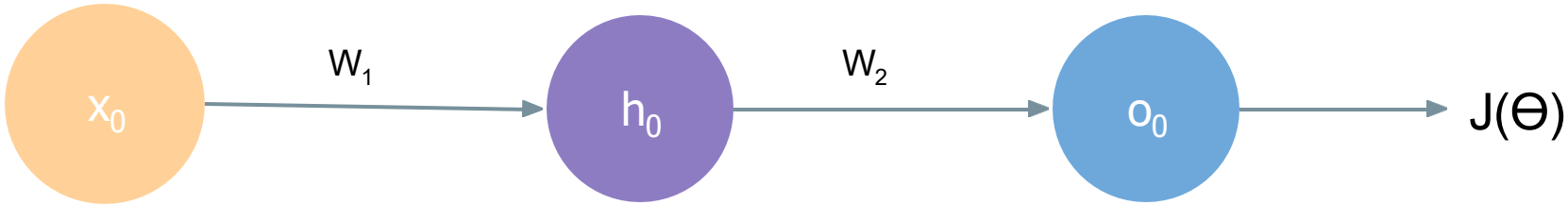


- How to Compute Gradient?

Calculating the Gradient: Backpropagation



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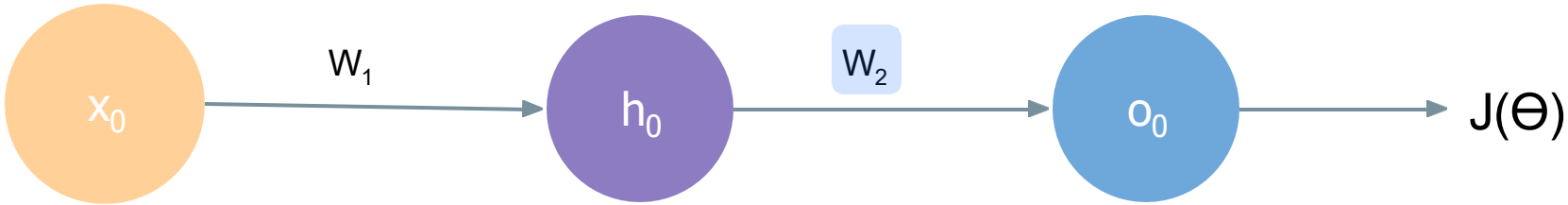
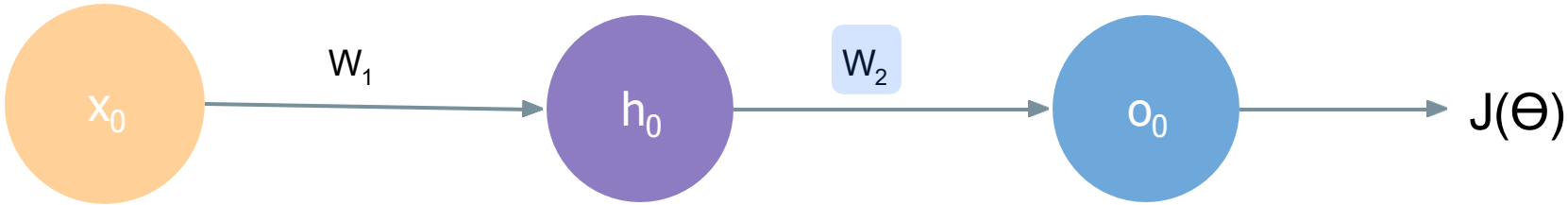


$$\frac{\partial J(\theta)}{\partial W_2} =$$

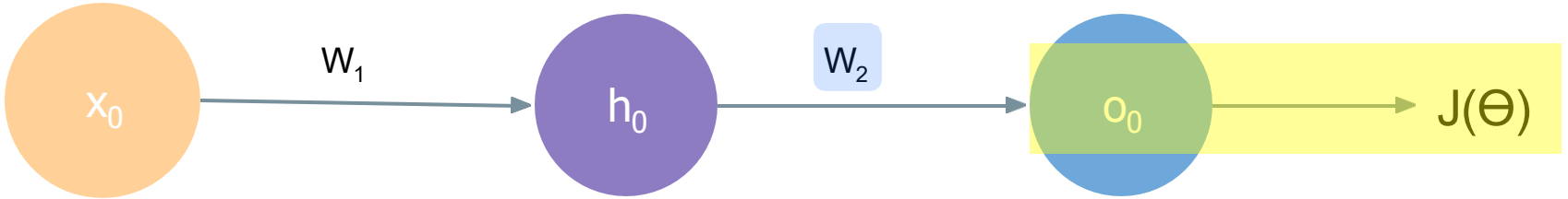
Calculating the Gradient: Backpropagation



Apply the chain rule

$$\frac{\partial J(\theta)}{\partial W_2} =$$

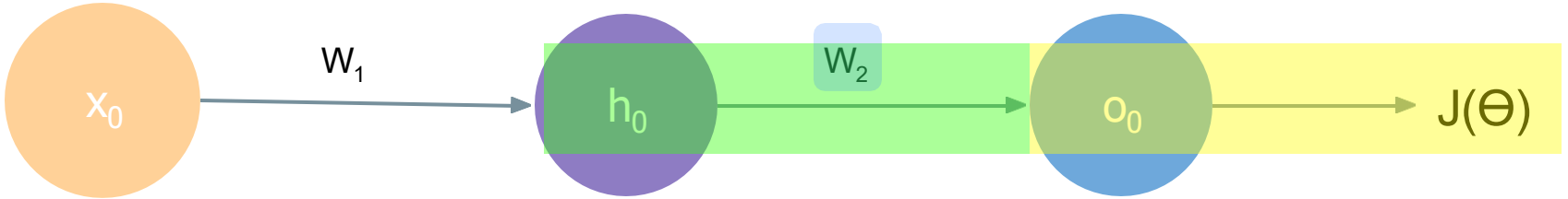
Calculating the Gradient: Backpropagation



Apply the chain rule

$$\frac{\partial J(\theta)}{\partial W_2} = \frac{\partial J(\theta)}{\partial o_0}$$

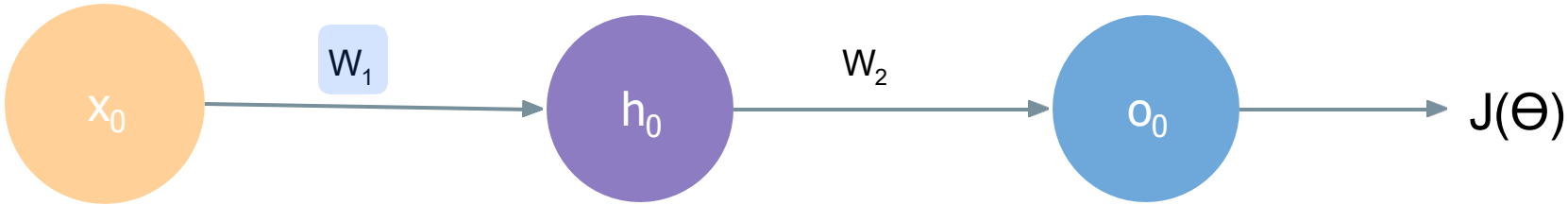
Calculating the Gradient: Backpropagation



Apply the chain rule

$$\frac{\partial J(\theta)}{\partial W_2} = \frac{\partial J(\theta)}{\partial o_0} * \frac{\partial o_0}{\partial W_2}$$

Calculating the Gradient: Backpropagation



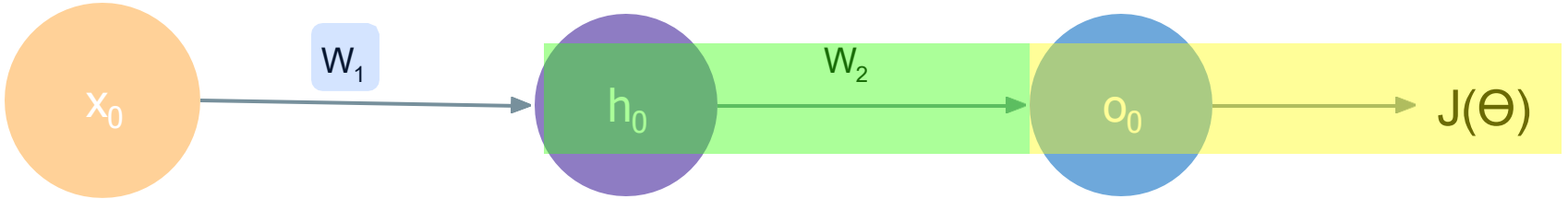
$$\frac{\partial J(\theta)}{\partial W_1} =$$

Calculating the Gradient: Backpropagation

Apply the chain rule

$$\frac{\partial J(\theta)}{\partial W_1} =$$

Calculating the Gradient: Backpropagation



Apply the chain rule

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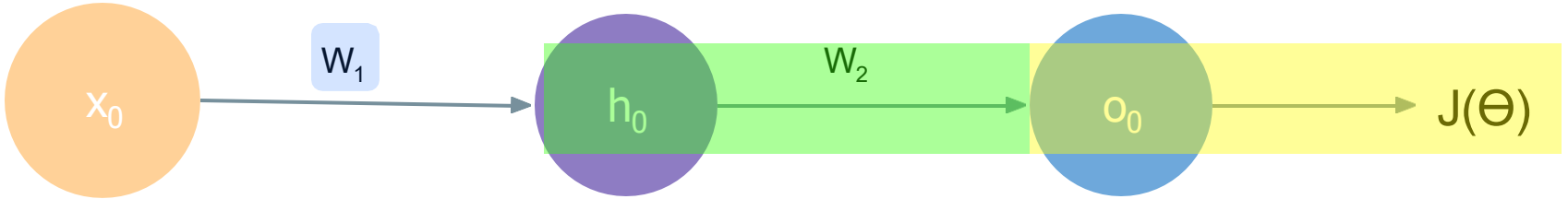

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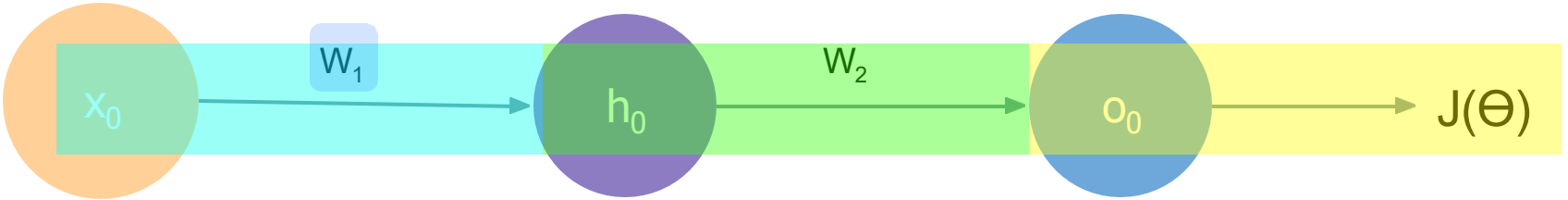

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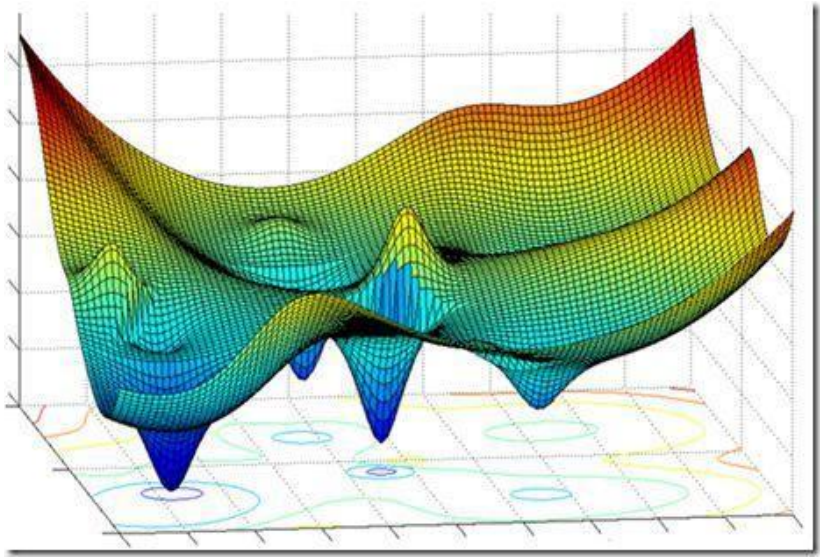
Apply the chain rule

$$\frac{\partial J(\theta)}{\partial W_1} = \frac{\partial J(\theta)}{\partial o_0} * \frac{\partial o_0}{\partial h_0} * \frac{\partial h_0}{\partial W_1}$$

Training Neural Networks In Practice

Loss function can be difficult to optimize



Loss function can be difficult to optimize

Update Rule: $$\theta := \theta - \eta \frac{\partial J(\theta)}{\partial \theta}$$

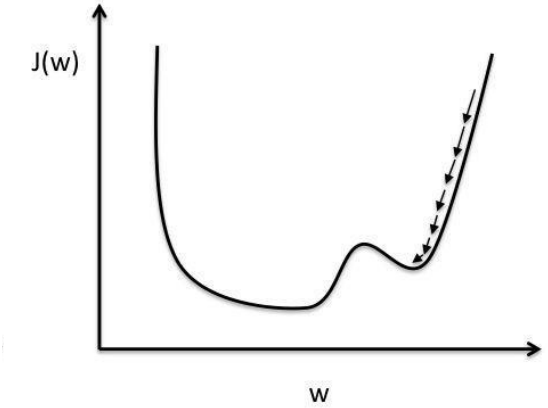
Loss function can be difficult to optimize

How to Choose Learning Rate?

Update Rule: $$\theta := \theta - \eta \frac{\partial J(\theta)}{\partial \theta}$$

Learning Rate & Optimization

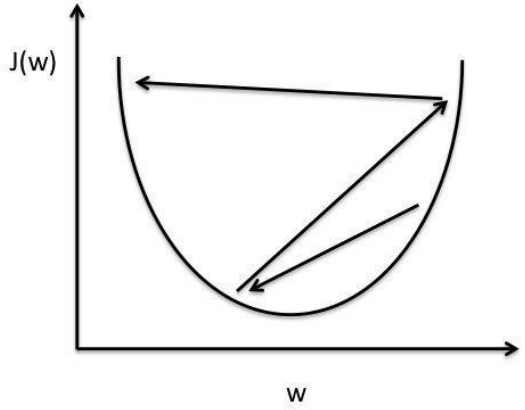
- Small Learning Rate



Small learning rate: Many iterations until convergence and trapping in local minima.

Learning Rate & Optimization

- Large learning rate



Large learning rate: Overshooting.

How to deal with this?

1. Try lots of different learning rates to see what is ‘just right’

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2. Do something smarter

How to deal with this?

1. Try lots of different learning rates to see what is ‘just right’
2. **Do something smarter : Adaptive Learning Rate**

Adaptive Learning Rate

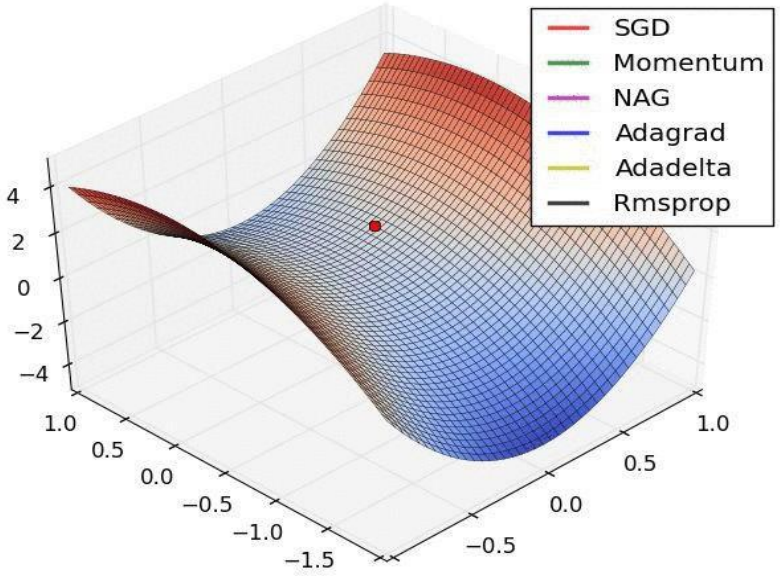
- Learning rate is no longer fixed
- Can be made larger or smaller depending on:
  - how large gradient is
  - how fast learning is happening
  - size of particular weights
  - etc

Adaptive Learning Rate Algorithms

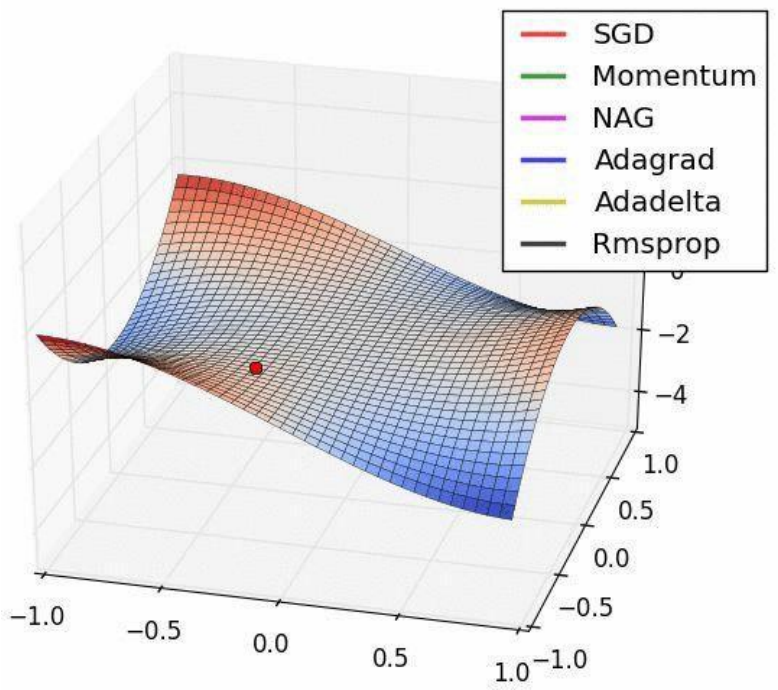
- ADAM
- Momentum
- NAG
- Adagrad
- Adadelta
- RMSProp

For details: check out http://sebastianruder.com/optimizing-gradient-descent/

Escaping Saddle Points



Escaping Saddle Points



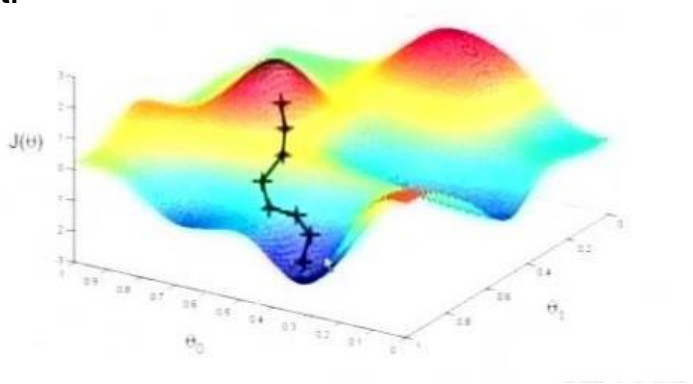
Training Neural Networks In Practice 2: MiniBatches

Why is it **Stochastic** Gradient Descent?

- Initialize θ randomly
- For N Epochs
  - For each training example (x, y):
    - Compute Loss Gradient: $\frac{\partial J(\theta)}{\partial \theta}$
    - Update θ with update rule:

$$\theta := \theta - \eta \frac{\partial J(\theta)}{\partial \theta}$$

Only an estimate of true gradient!



Minibatches Reduce Gradient Variance

- Initialize θ randomly
- For N Epochs
  - For each training **batch {(x0, y0), … , (x_B, y_B)}:**
    - Compute Loss Gradient: $\frac{\partial J(\theta)}{\partial \theta} = \frac{1}{B}\sum_{i}^{B}\frac{\partial J_i(\theta)}{\partial \theta}$
    - Update θ with update rule:

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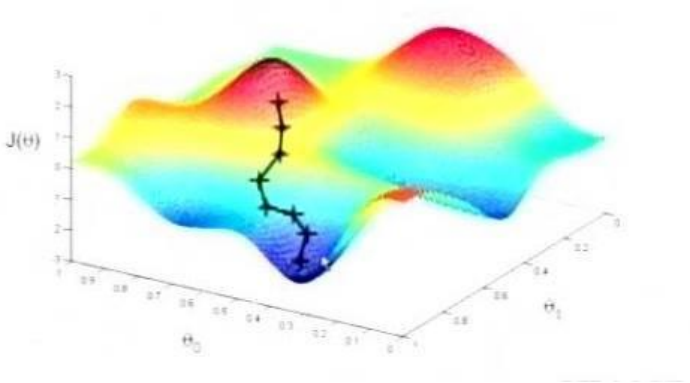
More accurate estimate!

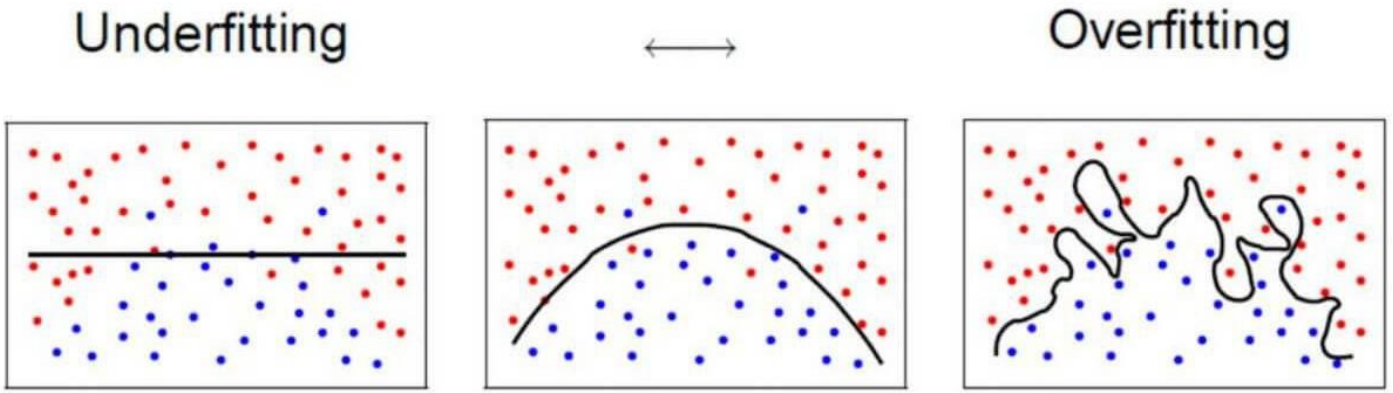


Advantages of Minibatches

- More accurate estimation of gradient
  - Smoother convergence
  - Allows for larger learning rates
- Minibatches lead to fast training!
  - Can parallelize computation + achieve significant speed increases on GPU’s

Training Neural Networks In Practice 3: Fighting Overfitting

The Problem of Overfitting

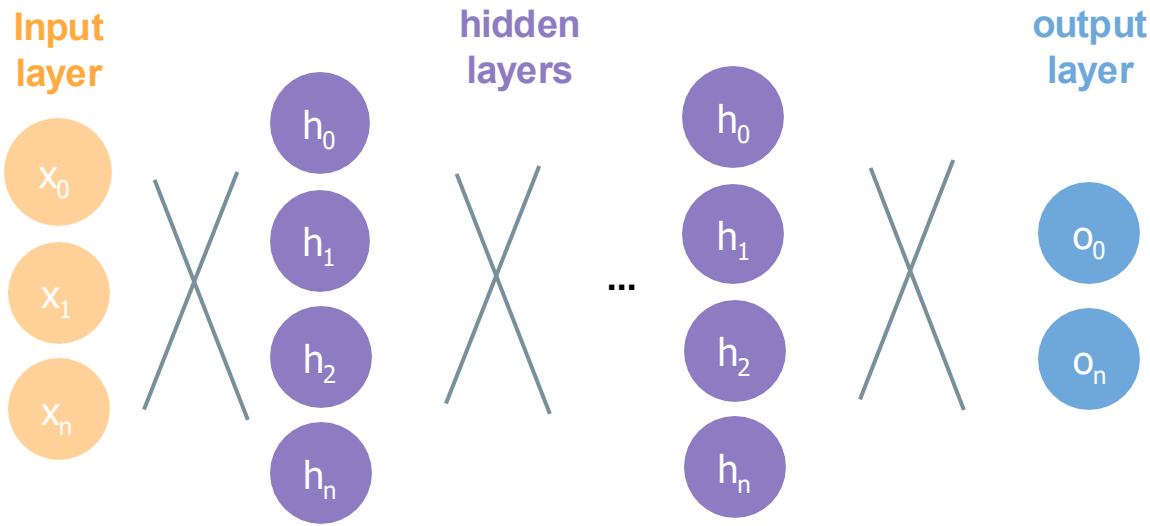


Regularization Techniques

1. Dropout
2. Early Stopping
3. Weight Regularization
4. ...many more

Regularization I: Dropout

- During training, randomly set some activations to 0



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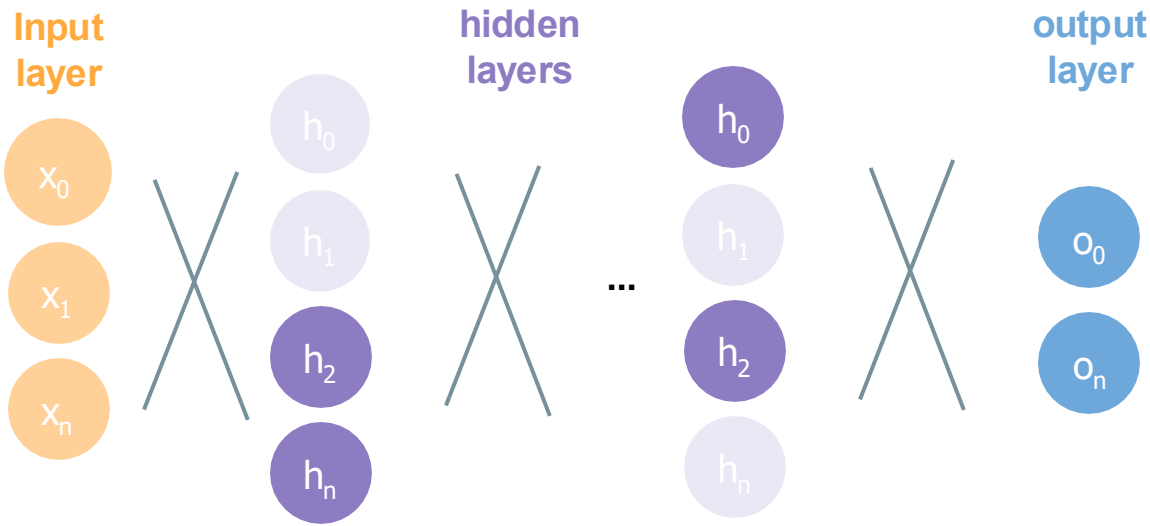
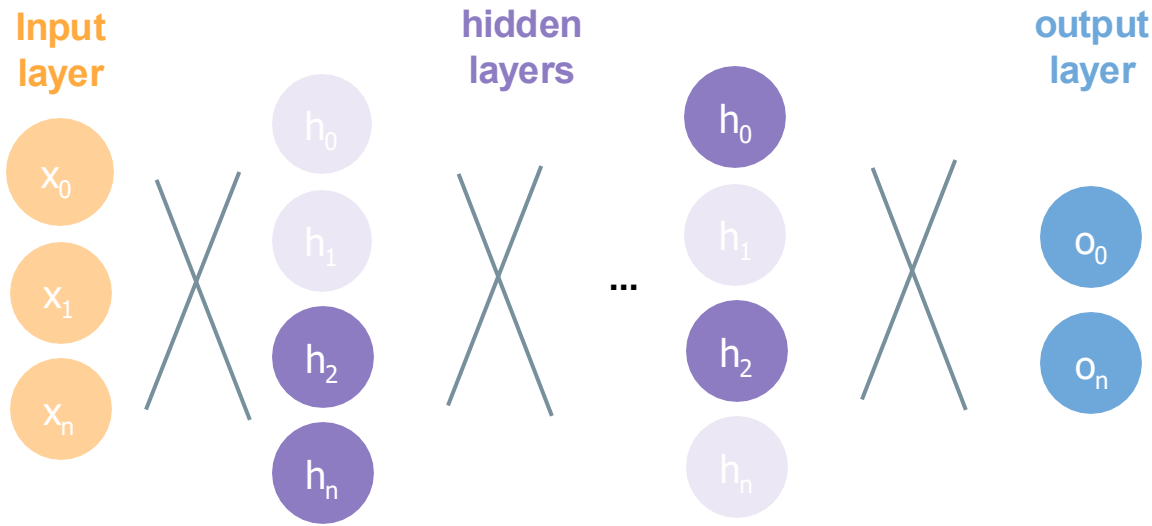


Regularization I: Dropout

- During training, randomly set some activations to 0
  - Typically ‘drop’ 50% of activations in layer
  - Forces network to not rely on any 1 node



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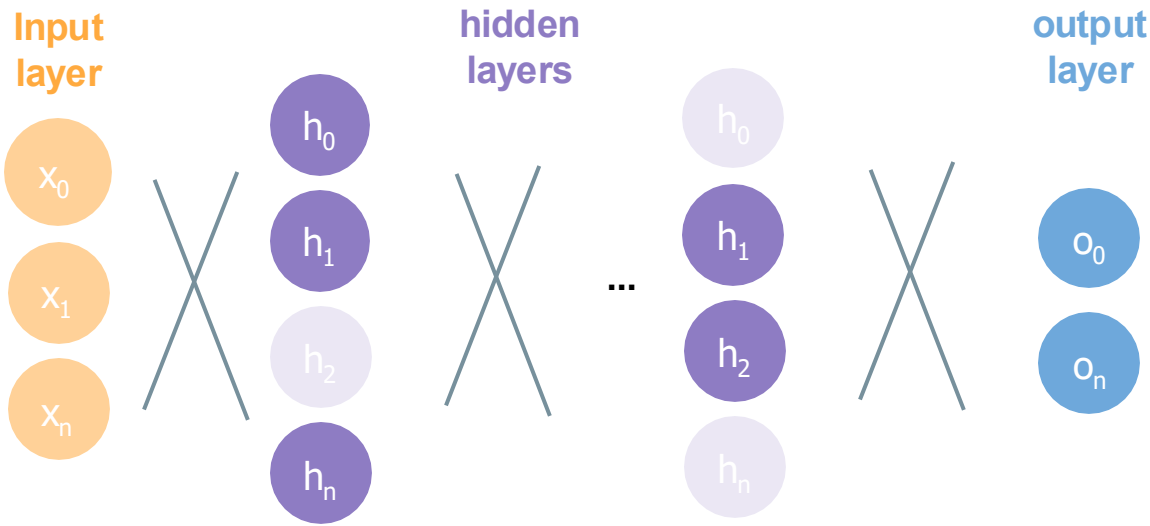

Regularization II: Early Stopping

- Don’t give the network time to overfit
- ...
- **Epoch 15:** Train: 85% Validation: 80%
- **Epoch 16:** Train: 87% Validation: 82%
- **Epoch 17:** Train: 90% Validation: 85%
- **Epoch 18:** Train: 95% Validation: 83%
- **Epoch 19:** Train: 97% Validation: 78%
- **Epoch 20:** Train: 98% Validation: 75%

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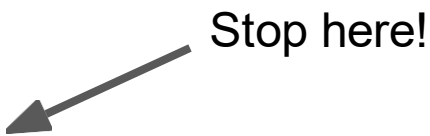

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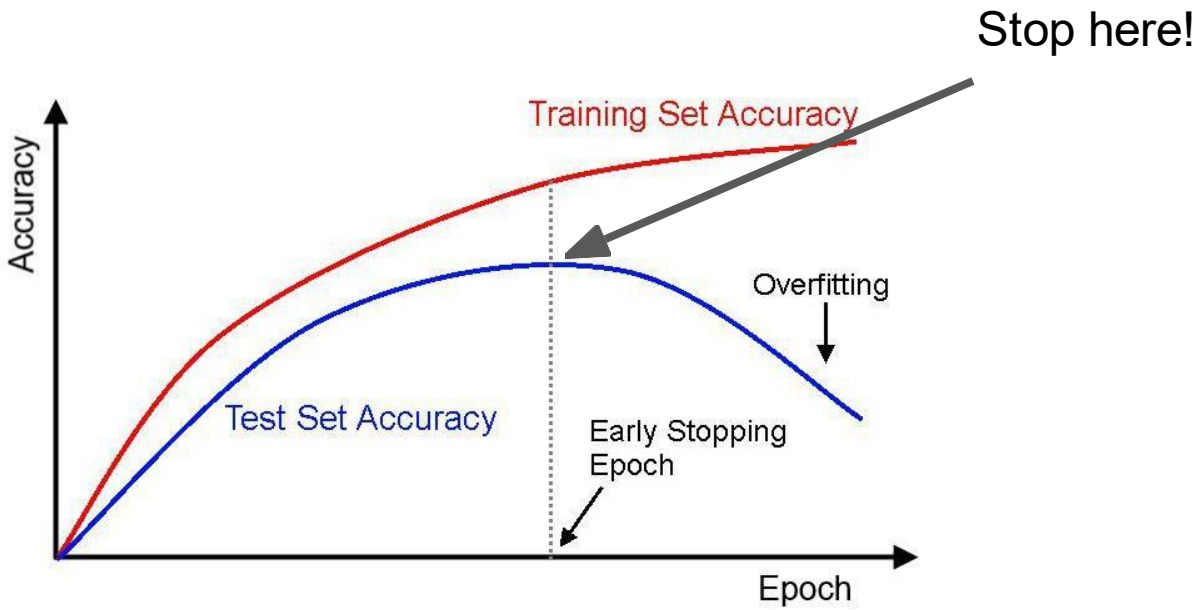

Regularization III: Weight Regularization

- Large weights typically mean model is overfitting
- Add the size of the weights to our loss function
- Perform well on task + keep weights small

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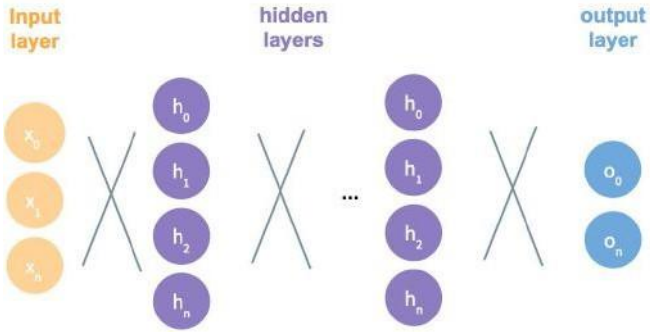
Core Fundamentals Review

- Perceptron Classifier
- Stacking Perceptrons to form neural networks
- How to formulate problems with neural networks
- Train neural networks with backpropagation
- Techniques for improving training of deep neural networks



References

- Hands on machine learning with scikit learn : Geron aurelien